TOPICS IN HIGH-DIMENSIONAL MANIFOLD TOPOLOGY

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Preface

These are personal lecture notes for **Prof. Yang Su's High-dimensional Geometry and Topology course** during the 2026 summer school of geometry and topology at Sichuan University.

Before reading, it is important to note that these notes are not endorsed by the lecturers and that I have modified them (often significantly) after each lecture. They are nowhere near accurate representations of what was actually lectured, and in particular, **all errors are almost surely mine**.

I would also like to thank the great power of AI (mainly ChatGPT 6-Astra) for helping me generate the TikZ code, create the beautiful images, and smooth my expression.

If you find typos or errors in this paper, please contact `12311501@mail.sustech.edu.cn`. Suggestions and corrections are welcome.

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1 Introduction to Manifold Topology

These notes assume familiarity with basic algebraic topology and smooth manifolds. We study classification, maps and symmetries in the topological and smooth categories. Two successive classification problems organize the course:

Homotopy types of finite complexes $\longrightarrow$ topological manifolds $\longrightarrow$ smooth manifolds.

The manifold realization problem. Given a finite complex K, is there a closed manifold M with $M \simeq K$? If so, how many manifold structures realize this homotopy type? Surgery theory addresses this problem.

The smoothing problem. Given a topological manifold, does it admit a smooth structure? If so, how many smooth structures does it admit, up to diffeomorphism? Exotic spheres provide fundamental examples.

The homotopy-rigidity question. The conjecture called the Hurewicz conjecture in the lecture asks whether every homotopy equivalence between closed manifolds is homotopic to a homeomorphism. This is false: $L^3(7,1)$ and $L^3(7,2)$ are homotopy equivalent but not homeomorphic. Counterexamples also exist among simply connected smooth manifolds.

Theorem 1.1 (Novikov)**.** *Rational Pontryagin classes are topological invariants. If $h : M \to N$ is a homeomorphism between smooth manifolds, then*

$$h^* p_i(TN)_{\mathbb{Q}} = p_i(TM)_{\mathbb{Q}}.$$

Consequently, if a homotopy equivalence $f : M \to N$ fails this equality, it cannot be homotopic to a homeomorphism. To rule out every homeomorphism, one needs an obstruction independent of the choice of f. For example, if $p_1(TM)_{\mathbb{Q}} \neq 0$ but $p_1(TN)_{\mathbb{Q}} = 0$, no homeomorphism exists. The sphere-bundle construction later in the notes has this property. Classes on different spaces must always be compared through an induced cohomology map.

Borel conjecture. If M is a closed aspherical manifold and N is a closed manifold, every homotopy equivalence $N \to M$ is conjectured to be homotopic to a homeomorphism.

Definition 1.2 (Aspherical)**.** A connected space X is aspherical if $\pi_i(X) = 0$ for every $i \geq 2$; its fundamental group need not vanish.

For a connected CW-complex, asphericity is equivalent to contractibility of the universal cover: covering maps preserve higher homotopy groups, and Whitehead's theorem applies to the simply connected universal cover. The same statement holds for manifolds, which have CW homotopy type.

Example 1.3. The circle and torus have contractible universal covers $\mathbb{R}$ and $\mathbb{R}^2$. A closed orientable surface of genus at least two has universal cover $\mathbb{H}^2$ and is also aspherical. In contrast, S^2 is not aspherical because $\pi_2(S^2) \cong \mathbb{Z}$.

Theorem 1.4 (Mostow rigidity)**.** *For closed hyperbolic manifolds M^n, N^n with $n \geq 3$, every homotopy equivalence $M \to N$ is homotopic to an isometry.*

Cartan–Hadamard implies that a complete manifold of nonpositive sectional curvature has contractible universal cover. This establishes asphericity; Mostow rigidity is a separate theorem. The lecture also mentions nonpositively curved and Lie-group-related examples in connection with Borel rigidity, but those extensions require their own hypotheses.

Generalized Poincaré conjecture. A closed manifold homotopy equivalent to S^n is called a homotopy sphere. The topological generalized Poincaré conjecture, now a theorem in every dimension, says that every such manifold is homeomorphic to S^n. Perelman's work resolved the three-dimensional case. In the smooth category, exotic 7-spheres already show that the analogous statement fails in general. The smooth four-dimensional question is a separate problem; the high-dimensional arguments below do not settle it.

Remark 1.5. For $n \geq 2$, a closed connected n-manifold is a homotopy sphere if and only if it is simply connected and has the integral homology of S^n.

High dimensions allow both more flexibility and more complexity. Every finitely presented group occurs as the fundamental group of a closed smooth manifold of any prescribed dimension $d \geq 4$. On the other hand, general position and the Whitney trick permit geometric cancellations under suitable dimension and fundamental-group hypotheses. These tools are central to the h-cobordism and surgery theorems.

Theorem 1.6 (Hurewicz)**.** *Let $n \geq 2$ and let X be a connected CW-complex with $\pi_i(X) = 0$ for $1 \leq i < n$. Then $\widetilde{H}_i(X; \mathbb{Z}) = 0$ for $i < n$, and*

$$\pi_n(X) \longrightarrow H_n(X; \mathbb{Z}), \qquad [f : S^n \to X] \longmapsto f_*[S^n]$$

is an isomorphism.

Theorem 1.7 (Whitehead)**.** *A map $f : X \to Y$ between connected CW-complexes is a homotopy equivalence if it induces isomorphisms on all homotopy groups (with compatible basepoints).*

Corollary 1.8 (Homological Whitehead theorem)**.** *A map between simply connected CW-complexes that induces isomorphisms on all integral homology groups is a homotopy equivalence.*

The requirement that a map induce the homology isomorphisms is essential; abstractly isomorphic homology groups alone do not suffice. We use these results repeatedly for manifolds and their boundary inclusions.

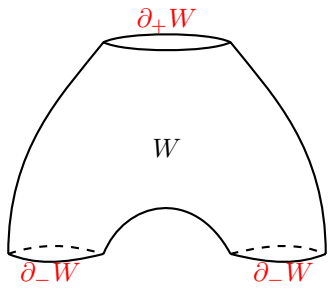
2 h-Cobordism Theorem and its Applications

Cobordism studies manifolds as boundaries. We state the smooth h-cobordism theorem and develop its applications to homotopy spheres and products with Euclidean spaces.

One can follow Milnor's *'Lectures on the h-cobordism theorem'*[Mil25] for all the details and tricks.

Definition 2.1 (Cobordism)**.** A cobordism W is a compact manifold with boundary, together with a decomposition of ∂W into two disjoint unions of boundary components $\partial_+ W$ and $\partial_- W$.

A visual picture of the cobordism can be seen as the following, where there is no constraint on the dimension of the W:



Definition 2.2 (h-cobordism)**.** We say a cobordism W is an h-cobordism if both $\partial_+ W \to W$ and $\partial_- W \to W$ are homotopy equivalences.

Theorem 2.3 (h-Cobordism Theorem (Smale))**.** *Let W^n be a simply connected smooth h-cobordism, $n \geq 6$. Then there exists a diffeomorphism $f : \partial_- W \times [0,1] \to W$ s.t. $f|_{\partial_- W} = id$. In particular, $f|_{\partial_- W \times \{1\}}$ identifies $\partial_- W$ diffeomorphically with $\partial_+ W$.*

This is a very important theorem in the field of high dimensional geometric topology, since this provides us with a possible way to construct diffeomorphism between high dimensional geometric objects with some good topological properties.

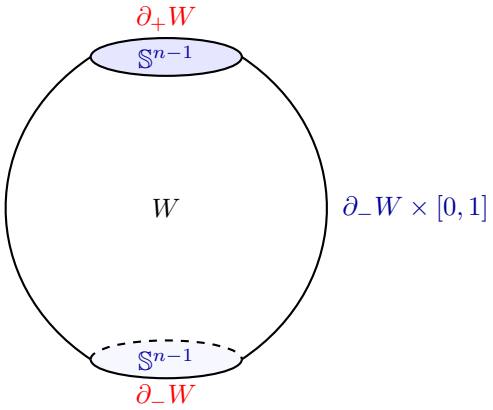
2.1 Proof of the Generalised Poincaré conjecture

Recall the statement of the GPC (Generalized Poincaré Conjecture) as follows:

Theorem 2.4 (Generalized Poincaré Conjecture)**.** *Let M^n be the closed $C^\infty-$manifold ($n \geq 6$) and $M^n \simeq \mathbb{S}^n$, then M^n is homeomorphic to $\mathbb{S}^n$.*

We prove this using the h-cobordism theorem.

Proof. We will give a picture first for some geometric intuitions with the definitions:



Choose disjoint smoothly embedded closed disks $D^n_+, D^n_- \subset M$ and remove their interiors:

$$W = M \setminus (\operatorname{int} D^n_+ \cup \operatorname{int} D^n_-), \qquad \partial_\pm W = \partial D^n_\pm.$$

We verify that both boundary inclusions are homotopy equivalences.

1. Van Kampen's theorem, applied with collars to the disk boundaries, gives $\pi_1(W) \cong \pi_1(M) = 0$, since $n \geq 3$.

2. Put $A = M \setminus \operatorname{int} D^n_+$. The Mayer–Vietoris sequence for $M = A \cup D^n_+$ shows that $\widetilde{H}_*(A; \mathbb{Z}) = 0$. In the only nontrivial high-degree part,

$$0 \longrightarrow H_n(M) \xrightarrow{\delta} H_{n-1}(\partial D^n_+) \longrightarrow H_{n-1}(A) \longrightarrow 0,$$

   δ sends the fundamental class of M to the boundary sphere's fundamental class, up to sign; hence it is an isomorphism $\mathbb{Z} \to \mathbb{Z}$. In degree zero, A is connected and $H_0(A) = \mathbb{Z}$.

3. Excision (using collars) and the long exact sequence of a pair give

$$H_k(W, \partial_- W; \mathbb{Z}) \cong H_k(A, D^n_-; \mathbb{Z}) = 0 \qquad \text{for every } k.$$

   Thus the inclusion $\partial_- W \hookrightarrow W$ induces homology isomorphisms. Both spaces are simply connected and have CW type, so the homological Whitehead theorem makes this inclusion a homotopy equivalence. Interchanging the two disks proves the same assertion for $\partial_+ W$.

The h-cobordism theorem therefore gives

$$W^n = M \setminus (\operatorname{int} D^n_+ \cup \operatorname{int} D^n_-) \underset{\text{diffeo.}}{\cong} \mathbb{S}^{n-1} \times [0,1].$$

The next idea is to glue the two disks back along the boundary components. Before doing so, we introduce the well-known Alexander trick.

Theorem 2.5 (Alexander Trick)**.** *Let $f : \mathbb{S}^{n-1} \to \mathbb{S}^{n-1}$ be a diffeomorphism, then there exists a homeomorphism $F : \mathbb{D}^n \to \mathbb{D}^n$, such that:*

- *F is a homeomorphism.*

- $F|_{\mathbb{S}^{n-1}} = f$.
- $F(x, r) = (f(x), r), r \neq 0$ *and* $F(0, 0) = 0$.

From the result, this trick gives a way to extend diffeomorphism on $\mathbb{S}^{n-1}$ to a homeomorphism on $\mathbb{D}^n$.

Now consider the diffeomorphism

$$f : W \longrightarrow \mathbb{S}^{n-1} \times [0, 1]$$

given by the h-cobordism theorem, relative to the parametrized lower boundary. Use that parametrization to identify the lower disk with the standard disk. The maps then glue to a homeomorphism $g : M \to S^n$ as follows:

$$g = \begin{cases} F_+ & x \in \mathbb{D}^n_+ \\ f & x \in W^n \\ id. & x \in \mathbb{D}^n_- \end{cases}$$

where F_+ is the Alexander extension of the upper boundary map. The lower boundary restriction is the identity in the chosen parametrizations.

$$M^n = \mathbb{D}^n_- \cup W^n \cup \mathbb{D}^n_+ \qquad \mathbb{S}^n = \mathbb{D}^n \cup (\mathbb{S}^{n-1} \times [0, 1]) \cup \mathbb{D}^n$$

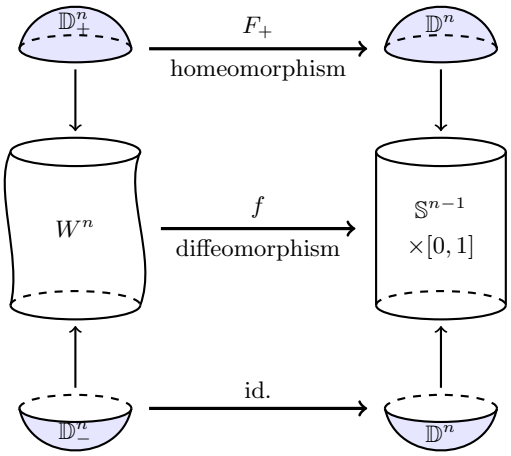


Figure 1: The piecewise maps are compatible with the boundary identifications and therefore glue to a homeomorphism $M^n \cong \mathbb{S}^n$.

Thus F_+, f, and id . are glued into a homeomorphism g

$$M^n \underset{\text{homeo.}}{\cong} \mathbb{S}^n.$$

□

Corollary 2.6 (Twisted-sphere presentation)**.** *Let Σ^n be a smooth homotopy sphere with*

$n \geq 6$. *There is a diffeomorphism* $\varphi : S^{n-1} \to S^{n-1}$ *such that*

$$\Sigma^n \cong D^n \cup_\varphi D^n.$$

Proof. Remove the interiors of two disjoint disks. The preceding argument identifies the remaining cobordism with $S^{n-1} \times [0,1]$. Absorb this collar into one disk; the remaining boundary identification is the diffeomorphism φ. The Alexander trick extends φ to a homeomorphism of the disk, but need not give a diffeomorphism at the center. □

Remark 2.7. h-cobordism gives us a very strong tool in creating a diffeomorphism between simply connected high dimensional objects.

We could have one more look on the application of the h-cobordism theorem to feel the great power of this theorem.

Proposition 2.8 (Browder)**.** *Let* M^n, N^n *be two closed, simply connected manifolds with dimension* $n \geq 5$. *If* $M \times \mathbb{R} \underset{\text{diffeo.}}{\cong} N \times \mathbb{R}$, *then* $M^n \cong N^n$.

Proof. Choose a diffeomorphism $f : M \times \mathbb{R} \to N \times \mathbb{R}$. Reverse the real coordinate if necessary so that it preserves the order of the two ends. Let

$$X_- = f\big(M \times (-\infty, 0]\big), \qquad X_+ = f\big(M \times [0, \infty)\big),$$

and let $Y_+ = N \times [0, \infty)$. Translate far enough in the positive real direction that $f(M \times \{0\})$ lies above $N \times \{0\}$ and $X_+ \subset Y_+$. The compact region between them is

$$W = X_- \cap Y_+.$$

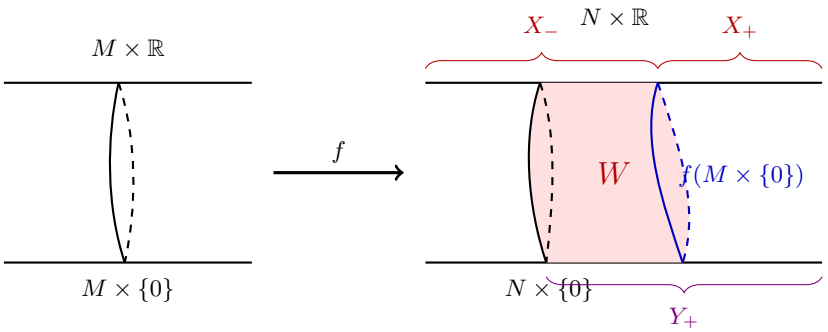


Figure 2: The hypersurfaces $N \times \{0\}$ and $f(M \times \{0\})$ bound the compact cobordism $W = X_- \cap Y_+$ inside $N \times \mathbb{R}$.

The next step for us is to show that the triple $(W, f(M \times \{0\}), N \times \{0\})$ is an h-cobordism.

$$W = Y_+ \cap X_-, \quad \text{and define: } \partial_- W = N \times \{0\},\ \partial_+ W = f(M \times \{0\})$$

First we show that the inclusion $\partial_+ W \hookrightarrow W$ induces homology isomorphisms, or equivalently that

$$H_*(W, \partial_+ W) \cong 0$$

Consider the excision property of the relative homology group, we have $H_*(W, \partial_+ W) \cong H_*(Y_+, X_+)$.

Use the long exact sequence of the triple $X_+ \subset Y_+ \subset N \times \mathbb{R}$, we get:

$$\begin{aligned} \cdots \longrightarrow H_{k+1}(N \times \mathbb{R}, Y_+) &\longrightarrow H_k(Y_+, X_+) \\ &\longrightarrow H_k(N \times \mathbb{R}, X_+) \\ &\longrightarrow H_k(N \times \mathbb{R}, Y_+) \longrightarrow \cdots . \end{aligned}$$

Since $Y_+ = N \times \mathbb{R}_+ \simeq N \times \mathbb{R}$, so $H_*(N \times \mathbb{R}, Y_+) \cong 0$, and $(N \times \mathbb{R}, X_+) \underset{\text{diffeo.}}{\cong} (M \times \mathbb{R}, M \times \mathbb{R}_+)$, so $H_*(N \times \mathbb{R}, X_+) \cong 0$. Consequently $H_*(Y_+, X_+) = 0$, and excision gives $H_*(W, \partial_+ W) = 0$.

To check simple connectivity, write $Y = N \times \mathbb{R}$ and $Y_- = N \times (-\infty, 0]$, van Kampen applied to $Y = Y_- \cup W \cup X_+$ (with collars) identifies $\pi_1(Y)$ with $\pi_1(W)$, because the two end pieces and the two gluing boundaries are simply connected. Hence W is simply connected. Poincaré–Lefschetz duality and the universal coefficient theorem now give $H_*(W, \partial_- W) = 0$ as well. Both inclusions are homotopy equivalences, so W is an h-cobordism. Its dimension is $n + 1 \geq 6$, and the h-cobordism theorem yields a diffeomorphism $M \cong N$. □

Cancellation also holds for $\mathbb{R}^2$; this requires a separate argument (see [BKS15]).

Proposition 2.9. *Let M^n, N^n be two closed, simply connected manifolds with dimension $n \geq 5$. If $M \times \mathbb{R}^2 \underset{\text{diffeo.}}{\cong} N \times \mathbb{R}^2$, then $M^n \cong N^n$.*

Cancellation fails for $\mathbb{R}^3$: the exotic 7-spheres in the lecture satisfy $\Sigma^7 \times \mathbb{R}^3 \cong S^7 \times \mathbb{R}^3$ despite $\Sigma^7 \ncong S^7$ diffeomorphically.

Theorem 2.10 (Hirsch)**.** *Let M^n, N^n be two closed, simply-connected, smooth manifolds, $n \geq 3$, and the mapping $f : M \to N$ is a stable tangential homotopy equivalence, i.e. $f^*(TN \oplus \mathbb{R}) \cong TM \oplus \mathbb{R}$. Then:*

$$M \times \mathbb{R}^{n+1} \cong N \times \mathbb{R}^{n+1}$$

Proof. We expand the tubular-neighborhood argument recorded in the lecture. By general position, the map $(f, 0) : M \to N \times \mathbb{R}^{n+1}$ is homotopic to a smooth embedding e, whose image can be placed in $N \times \operatorname{int} D^{n+1}$. Let ν_e be its normal bundle. Then

$$TM \oplus \nu_e \cong f^* TN \oplus \varepsilon^{n+1}.$$

The stable tangential hypothesis implies that ν_e is stably trivial. Its rank $n + 1$ exceeds $\dim M = n$, so stable bundle cancellation implies $\nu_e \cong \varepsilon^{n+1}$. Thus a closed tubular neighborhood is $U \cong M \times D^{n+1}$.

Set

$$V = N \times D^{n+1}, \qquad W = V \setminus \operatorname{int} U.$$

Its boundary components are $\partial_- W = \partial U \cong M \times S^n$ and $\partial_+ W = \partial V \cong N \times S^n$. Since $U \hookrightarrow V$ is a homotopy equivalence, excision gives

$$H_*(W, \partial_- W) \cong H_*(V, U) = 0.$$

The codimension of $e(M)$ is $n+1 \geq 4$. General position allows loops and their null-homotopies to avoid $e(M)$, so its complement, and therefore W, is simply connected. Duality gives $H_*(W, \partial_+ W) = 0$. Since both boundaries are also simply connected, their inclusions are homotopy equivalences.

Now $\dim W = 2n+1 \geq 7$, and the h-cobordism theorem identifies W with $(M \times S^n) \times [0,1]$ relative to ∂U. Gluing U back amounts to adding a collar:

$$N \times D^{n+1} = U \cup W \cong M \times D^{n+1}.$$

Taking interiors proves $N \times \mathbb{R}^{n+1} \cong M \times \mathbb{R}^{n+1}$. □

Example 2.11 (Homotopy spheres)**.** The lecture uses the Kervaire–Milnor fact that a smooth homotopy sphere Σ^n is stably parallelizable. Thus, for a homotopy equivalence $f : S^n \to \Sigma^n$,

$$f^*(T\Sigma^n \oplus \varepsilon^1) \cong TS^n \oplus \varepsilon^1 \cong \varepsilon^{n+1}.$$

For $n \geq 5$, Hirsch's theorem therefore gives

$$\Sigma^n \times \mathbb{R}^{n+1} \cong S^n \times \mathbb{R}^{n+1}.$$

2.2 A little Morse theory

Let $h : M^m \to \mathbb{R}$ be a Morse function and let $M_a = h^{-1}((-\infty, a])$ for a regular value a. On a compact band containing no critical points, the sublevel sets differ only by a collar. If the band $[a,b]$ contains exactly one critical point of index λ, then, after smoothing corners, M_b is obtained from M_a by attaching a λ-handle:

$$M_b \cong M_a \cup_\varphi (D^\lambda \times D^{m-\lambda}), \qquad \varphi : S^{\lambda-1} \times D^{m-\lambda} \hookrightarrow \partial M_a.$$

The local model is

$$h(x) = h(p) - x_1^2 - \cdots - x_\lambda^2 + x_{\lambda+1}^2 + \cdots + x_m^2.$$

The disk $D^\lambda \times \{0\}$ is the core of the handle; $\{0\} \times D^{m-\lambda}$ is its cocore.

Example 2.12 (The height function on a torus)**.** Choose a generic height function on T^2 with four critical points, ordered by height, of indices $0, 1, 1, 2$. Starting below the minimum, the successive sublevel sets are the empty set, a disk, an annulus, a once-punctured torus, and finally the closed torus. Thus the torus admits a handle decomposition with one 0-handle, two 1-handles, and one 2-handle, giving $\chi(T^2) = 1 - 2 + 1 = 0$.

For a cobordism $(W; M_-, M_+)$, a Morse function constant on each boundary gives a relative handle decomposition starting from $M_- \times [0,1]$. The condition $H_*(W, M_-) = 0$ is the algebraic starting point for the handle-cancellation argument behind the h-cobordism theorem. The geometric cancellations require the dimension and simple-connectivity hypotheses; the vanishing of relative homology alone is not sufficient.

3 Milnor's Exotic Sphere

This section explains Milnor's construction of exotic spheres.

Theorem 3.1 (Milnor's Exotic Sphere)**.** *There exist 7-manifolds Σ^7 that are homeomorphic but not diffeomorphic to the standard 7-sphere $\mathbb{S}^7$.*

Milnor's exotic 7-sphere arises from a surprisingly concrete source: sphere bundles over $\mathbb{S}^4$. The key point is that vector bundles over a sphere are obtained by gluing two trivial bundles over the two hemispheres. The gluing map, called the clutching function, contains all the information.

We will state many theorems about K-theory and characteristic classes without proof to keep the idea clear.

3.1 Vector bundles over spheres

Let ξ be a rank k real vector bundle over $\mathbb{S}^n$. Write

$$\mathbb{S}^n = \mathbb{D}^n_+ \cup_{\mathbb{S}^{n-1}} \mathbb{D}^n_-$$

for the decomposition into upper and lower hemispheres with equator $\mathbb{S}^{n-1}$. Since each disk is contractible, ξ is trivial on both hemispheres. Thus we can construct a vector bundle in the following way.

$$E(\xi) = (\mathbb{D}^n_+ \times \mathbb{R}^k) \cup_\alpha (\mathbb{D}^n_- \times \mathbb{R}^k),$$

where

$$\alpha : \mathbb{S}^{n-1} \longrightarrow GL_k(\mathbb{R})$$

is the transition function along the equator. After choosing a bundle metric, we may replace $GL_k(\mathbb{R})$ by $O(k)$.

Definition 3.2 (Clutching function)**.** The map

$$\alpha : \mathbb{S}^{n-1} \longrightarrow O(k)$$

used to glue the two trivial bundles over $\mathbb{D}^n_+$ and $\mathbb{D}^n_-$ is called the clutching function of ξ.

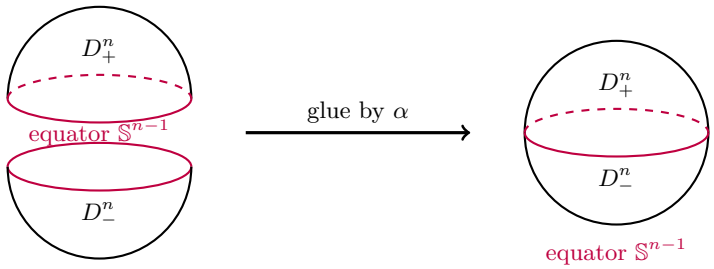


Figure 3: A vector bundle over a sphere is determined by a clutching function along the equator.

Clutching is a classification by homotopy classes, not by individual transition functions.

Proposition 3.3 (Oriented clutching classification)**.** *For $n \geq 2$, oriented rank k real bundles over S^n, up to orientation-preserving bundle isomorphism, correspond bijectively to*

$$\mathrm{Vect}_+^k(S^n) \longleftrightarrow \pi_{n-1}(SO(k)).$$

Homotopic clutching functions give isomorphic bundles; conversely, changes of trivialization over the two disks do not change this homotopy class.

The group structure used below is transported from the homotopy group. It is not the Whitney sum of rank k bundles, which would have rank $2k$. Forgetting orientation requires a further identification by orientation reversal; one should not identify the unoriented classification with the same group without this qualification.

Milnor studied oriented rank 4 real vector bundles over $\mathbb{S}^4$, the clutching bundles are classified by $\pi_3(SO(4))$. To compute the group's structure, we need the following facts:

Proposition 3.4. *A covering map $p : \left(\tilde{X}, \tilde{x}_0\right) \to (X, x_0)$ induces isomorphisms $p_* : \pi_n\left(\tilde{X}, \tilde{x}_0\right) \cong \pi_n(X, x_0)$ for all $n \geq 2$.*

Thus covering maps preserve higher homotopy groups.

Proposition 3.5. *There exists a double cover from $\mathbb{S}^3 \times \mathbb{S}^3 \to SO(4)$.*

Then we can compute the group.

$$\pi_3(SO(4)) \cong \pi_3(\mathbb{S}^3 \times \mathbb{S}^3) \cong \mathbb{Z} \oplus \mathbb{Z},$$

The group $\mathrm{Vect}_+^4(\mathbb{S}^4)$ therefore has two free generators. As with choosing a basis for a vector space, it remains to construct convenient generators from familiar canonical bundles. Readers may pause here and try to find them directly.

For a convenient basis, let $\tau = T\mathbb{S}^4$ be the tangent bundle and let $\eta_{\mathbb{H}}$ be the quaternionic Hopf line bundle over $\mathbb{S}^4 = \mathbb{H}P^1$. If u denotes the generator of $H^4(\mathbb{S}^4; \mathbb{Z})$, then

	e	p_1
$\tau = T\mathbb{S}^4$	$2u$	0
$\eta_{\mathbb{H}}$	u	$-2u$

In additive notation for the classification group, write

$$\xi_{a,b} = a\tau + b\eta_{\mathbb{H}}, \qquad a, b \in \mathbb{Z}.$$

Here addition takes place in $\pi_3(SO(4))$, and the characteristic numbers are additive in this clutching-group notation. Keeping the generator explicit,

$$e(\xi_{a,b}) = (2a + b)u, \qquad p_1(\xi_{a,b}) = -2bu.$$

Let

$$M_{a,b} = S(\xi_{a,b})$$

be the unit sphere bundle. It fits into a fibration

$$\mathbb{S}^3 \longrightarrow M_{a,b} \xrightarrow{\pi} \mathbb{S}^4.$$

$$\mathbb{S}^3 \longrightarrow M_{a,b} = S(\xi_{a,b}) \xrightarrow{\pi} \mathbb{S}^4$$

Figure 4: Milnor's manifolds are total spaces of $\mathbb{S}^3$-bundles over $\mathbb{S}^4$.

Theorem 3.6. *The total space $M_{a,b}$ is a homotopy 7-sphere if and only if $2a + b = \pm 1$.*

The following calculation uses the Gysin exact sequence of an oriented sphere bundle.

Proof. The long exact sequence of homotopy groups for

$$\mathbb{S}^3 \longrightarrow M_{a,b} \longrightarrow \mathbb{S}^4$$

gives $\pi_1(M_{a,b}) = 0$. The Gysin sequence for the oriented sphere bundle contains the map

$$H^0(\mathbb{S}^4) \xrightarrow{\cup e(\xi_{a,b})} H^4(\mathbb{S}^4).$$

If $e(\xi_{a,b}) = \pm u$, equivalently $2a + b = \pm 1$, this map is an isomorphism. The Gysin sequence then gives

$$H^*(M_{a,b}; \mathbb{Z}) \cong H^*(\mathbb{S}^7; \mathbb{Z}).$$

Conversely, exactness gives $H^4(M_{a,b}; \mathbb{Z}) \cong \mathbb{Z}/(2a + b)$, with $\mathbb{Z}/0 = \mathbb{Z}$. A homotopy sphere has vanishing fourth cohomology, so $2a+b = \pm 1$ is necessary. Since $M_{a,b}$ is simply connected, the Hurewicz and Whitehead theorems imply

$$M_{a,b} \simeq \mathbb{S}^7.$$

□

By the generalized Poincare conjecture, each such $M_{a,b}$ is homeomorphic to $\mathbb{S}^7$. The remaining question is whether it is diffeomorphic to the standard sphere.

3.2 Signature and Hirzebruch's signature theorem

The above construction and computation gives us the complete algebraic criterion ($2a+b = 1$ or -1) for whether $M_{a,b} \underset{\text{homeo.}}{\cong} \mathbb{S}^7$ or not.

Now we introduce one more tools for helping us give another criterion for whether $M_{a,b} \underset{\text{diffeo.}}{\cong} \mathbb{S}^7$ or not.

Definition 3.7 (Signature)**.** Let X^{4k} be a closed oriented smooth manifold. The middle-dimensional cup product defines a symmetric bilinear form

$$S_X : H^{2k}(X; \mathbb{R}) \times H^{2k}(X; \mathbb{R}) \longrightarrow \mathbb{R}, \qquad (\alpha, \beta) \longmapsto \langle \alpha \smile \beta, [X] \rangle.$$

The signature of X is

$$\text{sign}(X) = \#\{\text{positive eigenvalues of } S_X\} - \#\{\text{negative eigenvalues of } S_X\}.$$

For example,

$$\text{sign}(\mathbb{S}^{4k}) = 0, \qquad \text{sign}(\mathbb{CP}^{2k}) = 1.$$

Remark 3.8. Poincaré duality makes S_X nondegenerate, and Sylvester's law of inertia makes the numbers of positive and negative eigenvalues independent of the chosen basis. For a manifold W^{4k} with boundary, use the induced nonsingular form on

$$\operatorname{im}\big(H^{2k}(W, \partial W; \mathbb{R}) \to H^{2k}(W; \mathbb{R})\big).$$

Equivalently, quotient the relative intersection form by its radical.

Example 3.9. For $S^2 \times S^2$, the two factor classes give the intersection matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, so its signature is zero. The torus T^4 also has signature zero; its tangent bundle is trivial, and the signature formula below gives $p_1(T^4)/3 = 0$.

Theorem 3.10 (Hirzebruch signature theorem)**.** *If X^{4k} is a closed oriented smooth manifold, then*

$$\operatorname{sign}(X) = \big\langle L_k\big(p_1(TX), \dots, p_k(TX)\big), [X] \big\rangle,$$

where L_k is the degree $4k$ Hirzebruch L-polynomial. In dimension $4, 8$,

$$L_1 = \frac{1}{3} p_1 \qquad L_2 = \frac{7p_2 - p_1^2}{45}.$$

Now assume $2a + b = \pm 1$, so that $M_{a,b}$ is a homotopy sphere. Let

$$W_{a,b} = D(\xi_{a,b})$$

be the disk bundle of $\xi_{a,b}$, whose boundary is $M_{a,b}$. If $M_{a,b}$ were diffeomorphic to the standard $\mathbb{S}^7$, then gluing an 8-disk to $W_{a,b}$ would give a closed smooth 8-manifold

$$X_{a,b} = D(\xi_{a,b}) \cup_{M_{a,b}} \mathbb{D}^8.$$

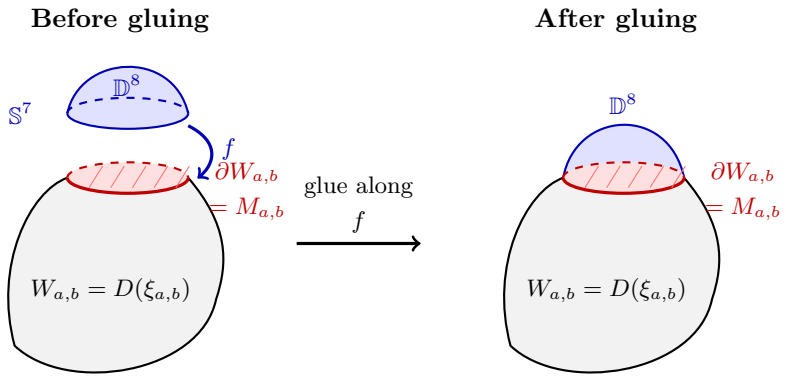


Figure 5: Attaching $\mathbb{D}^8$ to the disk bundle $D(\xi_{a,b})$ using a diffeomorphism $f : \mathbb{S}^7 \to M_{a,b}$ produces the closed smooth manifold $X_{a,b}$.

Mayer–Vietoris gives

$$H_i(X_{a,b}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & i = 0, 4, 8, \\ 0, & \text{otherwise.} \end{cases}$$

Put $s = 2a + b \in \{1, -1\}$ and orient the disk bundle using the orientations of the base and the fibers. The zero section has self-intersection s, so

$$\operatorname{sign}(X_{a,b}) = s.$$

Let $x \in H^4(X_{a,b}; \mathbb{Z})$ restrict to $\pi^* u$ on $W_{a,b}$. Since the zero section has self-intersection s, and $s^{-1} = s$, we have

$$\langle x^2, [X_{a,b}] \rangle = s.$$

A connection splits the tangent bundle of the disk bundle:

$$TW_{a,b} \cong \pi^*(TS^4 \oplus \xi_{a,b}), \qquad p_1(W_{a,b}) = -2b\pi^* u.$$

The restriction $H^4(X_{a,b}; \mathbb{Z}) \to H^4(W_{a,b}; \mathbb{Z})$ is an isomorphism. Consequently $p_1(X_{a,b}) = -2bx$ and

$$\langle p_1(X_{a,b})^2, [X_{a,b}] \rangle = 4sb^2.$$

Write $P_2 = \langle p_2(X_{a,b}), [X_{a,b}] \rangle \in \mathbb{Z}$. The signature theorem gives

$$s = \frac{7P_2 - 4sb^2}{45}, \qquad P_2 = \frac{s(45 + 4b^2)}{7}.$$

Thus a necessary condition for a diffeomorphism $M_{a,b} \cong S^7$ is

$$7 \mid 45 + 4b^2 \quad \Longleftrightarrow \quad b^2 \equiv 1 \pmod 7.$$

The signs in the signature and the Pontryagin square are linked by orientation; they cannot be chosen independently. This is an obstruction, not a sufficient criterion for being the standard sphere.

Corollary 3.11 (Milnor)**.** *If $2a + b = \pm 1$ and $b^2 \not\equiv 1 \pmod 7$, then $M_{a,b}$ is homeomorphic but not diffeomorphic to S^7. For example, $(a, b) = (-1, 3)$ gives $s = 1$ and $45 + 4b^2 = 81$, which is not divisible by 7.*

3.3 Spin and Rohlin's invariant

Before defining Milnor's λ invariant, we discuss the analogous Rohlin invariant in dimension 3.

3.3.1 Spin Structure

Definition 3.12 (Spin group)**.** Let Cl_n be the real Clifford algebra of $\mathbb{R}^n$, defined by the relations

$$vw + wv = -2\langle v, w \rangle \qquad (v, w \in \mathbb{R}^n).$$

Its even subalgebra is denoted by Cl_n^0. The spin group is

$$\mathrm{Spin}(n) = \left\{ v_1 v_2 \cdots v_{2r} \in \mathrm{Cl}_n^0 : r \geq 0,\ \|v_i\| = 1 \right\}.$$

Conjugation on $\mathbb{R}^n \subset \mathrm{Cl}_n$ defines a surjective homomorphism

$$\rho : \mathrm{Spin}(n) \longrightarrow SO(n), \qquad \rho(a)(x) = axa^{-1},$$

whose kernel is $\{\pm 1\}$. Thus there is a short exact sequence

$$1 \longrightarrow \{\pm 1\} \longrightarrow \mathrm{Spin}(n) \xrightarrow{\rho} SO(n) \longrightarrow 1.$$

For $n \geq 3$, this is the universal double cover of $SO(n)$.

The group $\mathrm{Spin}(n)$ therefore records the same infinitesimal rotations as $SO(n)$ while retaining additional global information about loops in $SO(n)$.

Definition 3.13 (Spin manifold)**.** Let W be a smooth manifold. We say that W is spin if

$$w_1(W) = 0, \qquad w_2(W) = 0,$$

where $w_i(W)$ are the Stiefel–Whitney classes. Equivalently, W is oriented and its principal $SO(n)$-frame bundle lifts to a principal $\mathrm{Spin}(n)$-bundle.

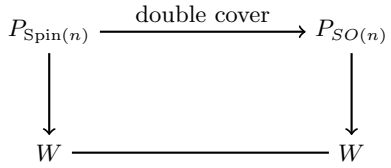


Figure 6: A spin structure is a lift of the oriented frame bundle through $\mathrm{Spin}(n) \to SO(n)$.

For $n \geq 3$, the covering above implies

$$\pi_1(SO(n)) \cong \mathbb{Z}/2.$$

Some useful low-dimensional identifications are

$$\mathrm{Spin}(2) \cong \mathbb{S}^1, \qquad \mathrm{Spin}(3) \cong SU(2) \cong \mathbb{S}^3, \qquad \mathrm{Spin}(4) \cong SU(2) \times SU(2).$$

In particular, $SO(3) \cong \mathbb{RP}^3$ and $\mathbb{S}^3 \to \mathbb{RP}^3$ is its universal double cover.

3.3.2 Rohlin Invariant

Theorem 3.14 (Rohlin)**.** *If X^4 is a closed smooth spin manifold, then*

$$16 \mid \mathrm{sign}(X).$$

To define an invariant for a general oriented 3-manifold, one must fix a spin structure on its boundary.

Definition 3.15 (Rohlin invariant)**.** Let $(M^3, \mathfrak{s})$ be a closed oriented spin 3-manifold. Choose a compact smooth spin filling $(W^4, \mathfrak{t})$ whose boundary spin structure is $\mathfrak{s}$ (such

a filling exists). Set

$$R(M,\mathfrak{s}) = \frac{\operatorname{sign}(W)}{16} \pmod{\mathbb{Z}} \in \mathbb{Q}/\mathbb{Z}.$$

Independence of the filling. Two fillings inducing the same boundary spin structure glue to a closed smooth spin manifold $X = W \cup_M (-W')$. Novikov additivity and Rohlin's theorem imply

$$\operatorname{sign}(W) - \operatorname{sign}(W') = \operatorname{sign}(X) \in 16\mathbb{Z}.$$

Therefore the displayed class is independent of the filling. An integral homology 3-sphere has a unique spin structure, so in that case one may write $R(M)$. □

3.4 Milnor's λ-invariant

The lecture's construction of λ only needs an oriented filling. Every oriented homotopy 7-sphere bounds a compact oriented smooth 8-manifold, since $\Omega_7^{SO} = 0$.

The relative Pontryagin square. Let $\partial W = \Sigma$, where Σ is a homotopy 7-sphere. The long exact sequence gives an isomorphism

$$j : H^4(W, \Sigma; \mathbb{Z}) \xrightarrow{\cong} H^4(W; \mathbb{Z}),$$

because $H^3(\Sigma) = H^4(\Sigma) = 0$. Define

$$\bar{p}_1 = j^{-1}(p_1(TW)), \qquad q(W) = \langle \bar{p}_1 \smile p_1(TW), [W, \Sigma] \rangle.$$

This specifies the relative cohomology interpretation of the shorthand $p_1(W)^2[W, \Sigma]$.

Definition 3.16 (The λ-invariant)**.** With the normalization used in the lecture, put

$$\lambda(\Sigma) = \frac{45 \operatorname{sign}(W) + q(W)}{7} \pmod{\mathbb{Z}} \in \tfrac{1}{7}\mathbb{Z}/\mathbb{Z}.$$

If W' is another oriented filling, form $X = W \cup_\Sigma (-W')$. The vanishing of the middle cohomology of Σ makes the relative Pontryagin squares additive under this gluing. The signature theorem then gives

$$\begin{aligned} 45\big(\operatorname{sign}(W) - \operatorname{sign}(W')\big) + q(W) - q(W') &= 45 \operatorname{sign}(X) + \langle p_1(X)^2, [X] \rangle \\ &= 7\langle p_2(X), [X] \rangle. \end{aligned}$$

Hence λ is independent of the filling modulo $\mathbb{Z}$. It changes sign under orientation reversal, is additive under connected sum, and vanishes on the standard sphere (use D^8).

Example 3.17 (Milnor's sphere bundles)**.** For $s = 2a + b = \pm 1$ and $W_{a,b} = D(\xi_{a,b})$, the relative intersection calculation above gives

$$\operatorname{sign}(W_{a,b}) = s, \qquad q(W_{a,b}) = 4sb^2,$$

so

$$\boxed{\lambda(M_{a,b}) = \frac{s(45 + 4b^2)}{7} \pmod{\mathbb{Z}}.}$$

For $(-1,3)$ this is $81/7 \equiv 4/7$, which detects an exotic sphere. The group $\frac{1}{7}\mathbb{Z}/\mathbb{Z}$ is identified with $\mathbb{Z}/7$ by $k/7 \mapsto k \pmod 7$; a fraction in $\mathbb{Q}/\mathbb{Z}$ and an integer residue use different normalizations.

Corollary 3.18 (A nonsmoothable topological manifold)**.** *Suppose $s = 2a + b = \pm 1$ and $7 \nmid 45 + 4b^2$. Glue D^8 to $D(\xi_{a,b})$ by a boundary homeomorphism. The resulting closed topological 8-manifold N admits no smooth structure.*

Proof. The topological intersection form has signature s. The restriction in degree four to the disk bundle is an isomorphism. Naturality and Novikov's topological invariance of rational Pontryagin classes force any proposed smoothing to have $p_1 = -2bx$, where $x^2[N] = s$. Hirzebruch's theorem would therefore force the integer Pontryagin number $p_2[N]$ to equal $s(45 + 4b^2)/7$, a contradiction. □

Remark 3.19. The hypothesis is the nonvanishing of this specific obstruction. Exoticness of the boundary alone does not imply it: λ is not a complete invariant of homotopy 7-spheres.

Hopf invariant one. The lecture also points toward sphere bundles $S^{r-1} \to \Sigma^{2r-1} \to S^r$ with Euler number ± 1. The associated Thom space has a two-cell description whose attaching map has Hopf invariant ± 1. Adams's Hopf invariant one theorem restricts r to $1, 2, 4, 8$; the complex, quaternionic, and octonionic cases have bases S^2, S^4, S^8. The S^7-bundle over S^8 case leads to homotopy 15-spheres.

Remark 3.20. A Fun fact in the history is that, although the example here are so closed to construct an non-smoothable topological manifold, the first example for topological manifold that admits no smooth structure is given by Kervaire constructing a 10 dimension manifold in his paper '*A manifold which does not admit any differentiable structure*'[Ker60] in 1960.

3.5 The Eells–Kuiper μ-invariant

The finding of Milnor's exotic sphere inspires people to ask further questions, how many different smooth structures on the $\mathbb{S}^n$.

Definition 3.21 (The group of homotopy spheres)**.** For $n \geq 5$, let Θ_n be the set of oriented smooth homotopy n-spheres up to orientation-preserving diffeomorphism. Connected sum gives an abelian group, with identity $[S^n]$ and inverse $[-\Sigma]$. Equivalently, one may use oriented h-cobordism classes in these dimensions.

Commutativity and associativity follow from the geometric connected-sum construction. The inverse assertion uses the high-dimensional h-cobordism theorem; it is not merely a formal property of reversing orientation.

The λ-invariant only takes seven possible values, whereas $\Theta_7 \cong \mathbb{Z}/28$. It cannot be a complete invariant. For a homotopy 7-sphere, choose a smooth spin filling W and use the relative square $q(W)$ defined above. The Eells–Kuiper invariant is normalized by

$$\mu(\Sigma) = \frac{q(W) - 4\,\mathrm{sign}(W)}{2^7 \cdot 7} \pmod{\mathbb{Z}}.$$

For a closed spin 8-manifold X, the signature and $\widehat{A}$ formulas give

$$\frac{p_1(X)^2[X] - 4\,\mathrm{sign}(X)}{896} = \widehat{A}(X) \in \mathbb{Z},$$

which proves independence of the spin filling. For homotopy 7-spheres,

$$\mu : \Theta_7 \xrightarrow{\cong} \tfrac{1}{28}\mathbb{Z}/\mathbb{Z}.$$

If one instead wants values in $\mathbb{Z}/28$, send $k/28 \pmod{\mathbb{Z}}$ to $k \pmod{28}$; this explicitly states the normalization. For the disk bundles in this section, $H^2(W_{a,b}; \mathbb{Z}/2) = 0$, so they are spin, and

$$\mu(M_{a,b}) = \frac{s(b^2-1)}{224} \pmod{\mathbb{Z}}.$$

For instance, $\mu(M_{-1,3}) = 1/28$ with our orientation convention. See [EK62] for the invariant and its general setting. The classification of Θ_7 will reappear in the surgery-theoretic discussion.

4 Counterexamples to the Hurewicz Conjecture

Recall the conjecture in the introduction section: we want to construct a smooth manifold M such that

$$M \simeq \mathbb{S}^4 \times \mathbb{S}^5,$$

but M is not homeomorphic to $\mathbb{S}^4 \times \mathbb{S}^5$. The construction uses sphere bundles over $\mathbb{S}^4$ and the stable J-homomorphism.

4.1 Sphere bundles over $\mathbb{S}^4$

Let $\xi_a \to \mathbb{S}^4$ be an oriented rank 6 vector bundle, and let

$$M_a = S(\xi_a)$$

be its unit sphere bundle. Thus M_a fits into a fibration

$$\mathbb{S}^5 \longrightarrow M_a \xrightarrow{\pi} \mathbb{S}^4.$$

$$\mathbb{S}^5 \xrightarrow{\text{incl.}} M_a = S(\xi_a) \xrightarrow{\pi} \mathbb{S}^4$$

Figure 7: The sphere bundle used to construct the counterexample.

Oriented rank 6 bundles over $\mathbb{S}^4$ are classified by clutching functions. Since the equator of $\mathbb{S}^4$ is $\mathbb{S}^3$, we have

$$\left\{\text{oriented rank 6 vector bundles over } \mathbb{S}^4\right\} \longleftrightarrow \pi_3(SO(6)) \cong \mathbb{Z}.$$

The integer $a \in \mathbb{Z}$ denotes the class of ξ_a. Choose the generator so that $p_1(\xi_1) = 2u$, where u is the positive generator of $H^4(\mathbb{S}^4; \mathbb{Z})$.

Stabilization is given by adding a trivial line bundle:

$$\xi_a \longmapsto \xi_a \oplus \mathbb{R}.$$

On clutching functions, this is induced by the standard inclusion

$$SO(6) \hookrightarrow SO(7), \qquad A \longmapsto \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

The inclusions $SO(k) \hookrightarrow SO(k+1)$ form the stable special orthogonal group

$$SO = \varinjlim_k SO(k).$$

They also induce stabilization maps

$$\pi_3(SO(k)) \longrightarrow \pi_3(SO(k+1)).$$

These maps are isomorphisms once k is sufficiently large; degree 3 is already stable for $SO(6)$. Consequently,

$$\pi_3(SO(6)) \xrightarrow{\cong} \pi_3(SO) = \varinjlim_k \pi_3(SO(k)) \cong \mathbb{Z}.$$

Thus stabilization does not lose any information about the bundle ξ_a.

We now explain the connection with real K-theory. Two real vector bundles ξ and η over a space X are called *stably isomorphic* if there exist trivial bundles such that

$$\xi \oplus \mathbb{R}^r \cong \eta \oplus \mathbb{R}^s.$$

The reduced real K-group $\widetilde{KO}(X)$ records these stable bundle classes after subtracting the rank. For a sphere, a rank k bundle is described by a clutching function

$$\alpha : \mathbb{S}^{n-1} \longrightarrow O(k).$$

Adding a trivial line bundle replaces α by its composite with $O(k) \hookrightarrow O(k+1)$. Therefore, after stabilization, clutching functions take values in

$$O = \varinjlim_k O(k),$$

and the clutching construction gives an isomorphism

$$\widetilde{KO}(\mathbb{S}^n) \cong [\mathbb{S}^{n-1}, O] = \pi_{n-1}(O).$$

For $n = 4$, this specializes to

$$\widetilde{KO}(\mathbb{S}^4) \cong \pi_3(O) = \pi_3(SO) \cong \mathbb{Z},$$

where $\pi_3(O) = \pi_3(SO)$ because positive-dimensional homotopy groups based at the identity only see the identity component of O. By Bott periodicity these groups are 8-periodic:

$n \bmod 8$	0	1	2	3	4	5	6	7
$\widetilde{KO}(\mathbb{S}^n)$	$\mathbb{Z}$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0

For comparison, complex K-theory is 2-periodic:

$$\widetilde{K}(\mathbb{S}^n) \cong \pi_{n-1}(U), \qquad \begin{array}{c|cc} n \bmod 2 & 0 & 1 \\ \hline \widetilde{K}(\mathbb{S}^n) & \mathbb{Z} & 0 \end{array}$$

The corresponding stable homotopy groups of the orthogonal group are

$m \bmod 8$	0	1	2	3	4	5	6	7
$\pi_m(O)$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

In the present construction, only the entry $\pi_3(SO) \cong \mathbb{Z}$ is needed.

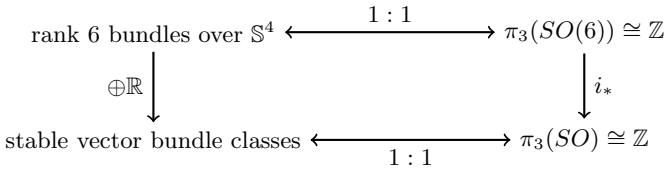


Figure 8: Classification of the vector bundles and their stable classes.

4.2 Fiber homotopy equivalence and the J-homomorphism

Definition 4.1 (Fiber homotopy equivalence)**.** Two sphere bundles $S(\xi) \to B$ and $S(\xi') \to B$ are fiber homotopy equivalent if there is a map

$$h : S(\xi) \longrightarrow S(\xi')$$

over B such that each restriction $h_b : S(\xi)_b \to S(\xi')_b$ is a homotopy equivalence of fibers.

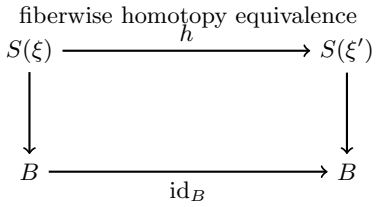


Figure 9: A fiber homotopy equivalence is a homotopy equivalence on each fiber and covers the identity on the base.

Let $G(k)$ denote the topological monoid of all self homotopy equivalences of $\mathbb{S}^{k-1}$, that is, maps of degree ± 1. Its identity component is

$$SG(k) = \mathrm{Map}_{\deg 1}(\mathbb{S}^{k-1}, \mathbb{S}^{k-1}).$$

For $n \geq 2$, oriented spherical fibrations over $\mathbb{S}^n$ are classified up to orientation-preserving fiber homotopy equivalence by $\pi_{n-1}(SG(k))$. Positive homotopy groups of $G(k)$ are based at the identity and equal those of $SG(k)$. The action of $O(k)$ on the sphere gives $O(k) \to G(k)$; its restriction sends $SO(k)$ to $SG(k)$. Suspension gives stabilization and, in positive degrees, the stable J-homomorphism

$$J_m : \pi_m(O) \longrightarrow \pi_m(G) \cong \pi_m^S, \qquad m > 0.$$

By Serre's finiteness theorem, π_m^S is finite for $m > 0$. The degree-zero group $\pi_0^S \cong \mathbb{Z}$ is excluded here.

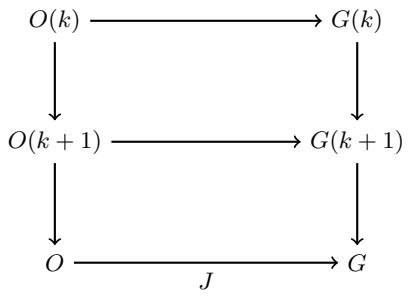


Figure 10: The stable J-homomorphism is induced by viewing orthogonal transformations as self homotopy equivalences of spheres.

Theorem 4.2 (Adams' image of J)**.** *For $m > 0$, the stable map $J_m : \pi_m(O) \to \pi_m^S$ is injective unless $m = 4k - 1$. In degree $4k - 1$, its image is cyclic of order*

$$\operatorname{den}\left(\frac{|B_{2k}|}{4k}\right),$$

where the fraction is reduced and B_{2k} is the classical Bernoulli number. In particular, the case needed below is as follows. The stable J-homomorphism in degree 3 is the reduction map

$$J : \pi_3(SO) \cong \mathbb{Z} \longrightarrow \pi_3^S \cong \mathbb{Z}/24.$$

Hence

$$\ker J = 24\mathbb{Z}.$$

4.3 The counterexample

Choose a nonzero integer

$$a \in \ker J = 24\mathbb{Z}.$$

We must check that stable triviality detects the actual $\mathbb{S}^5$-fibration in this dimension. Evaluation at a basepoint gives a fibration

$$\operatorname{Map}_*(\mathbb{S}^5, \mathbb{S}^5)_{\deg 1} \longrightarrow SG(6) \longrightarrow \mathbb{S}^5.$$

Since $\pi_3(\mathbb{S}^5) = \pi_4(\mathbb{S}^5) = 0$, its homotopy sequence and Freudenthal suspension give

$$\pi_3(SG(6)) \cong \pi_8(\mathbb{S}^5) \xrightarrow{\cong} \pi_3^S \cong \mathbb{Z}/24.$$

Thus the unstable clutching class vanishes whenever $J(a) = 0$. Consequently $M_a = S(\xi_a)$ is fiber homotopy equivalent to

$$M_0 = S(\xi_0) = \mathbb{S}^4 \times \mathbb{S}^5.$$

In particular,

$$M_a \simeq \mathbb{S}^4 \times \mathbb{S}^5.$$

It remains to show that M_a is not homeomorphic to $\mathbb{S}^4 \times \mathbb{S}^5$. We use Pontryagin classes.

The Gysin sequence for the sphere bundle $\mathbb{S}^5 \to M_a \xrightarrow{\pi} \mathbb{S}^4$ is

$$\cdots \longrightarrow H^{i-6}(\mathbb{S}^4) \xrightarrow{\cup e(\xi_a)} H^i(\mathbb{S}^4) \xrightarrow{\pi^*} H^i(M_a) \longrightarrow H^{i-5}(\mathbb{S}^4) \longrightarrow \cdots .$$

Because $e(\xi_a) \in H^6(\mathbb{S}^4) = 0$, the Gysin sequence gives

$$H^*(M_a) \cong H^*(\mathbb{S}^4 \times \mathbb{S}^5),$$

as graded abelian groups. To identify the ring, choose generators $x \in H^4(M_a;\mathbb{Z})$ and $y \in H^5(M_a;\mathbb{Z})$. Poincaré duality allows their signs to be chosen so that xy is the positive generator of $H^9(M_a;\mathbb{Z})$; their squares vanish for dimensional reasons. This gives the displayed isomorphism as rings as well. In particular,

$$\pi^* : H^4(\mathbb{S}^4) \xrightarrow{\cong} H^4(M_a).$$

The vertical tangent bundle satisfies

$$T_\pi M_a \oplus \mathbb{R} \cong \pi^* \xi_a.$$

Therefore

$$TM_a \oplus \mathbb{R} \cong T_\pi M_a \oplus \pi^* T\mathbb{S}^4 \oplus \mathbb{R} \cong \pi^*(\xi_a \oplus T\mathbb{S}^4).$$

$$\begin{array}{ccc} TM_a \oplus \mathbb{R} & \xrightarrow{\text{split tangent directions}} & T_\pi M_a \oplus \pi^* T\mathbb{S}^4 \oplus \mathbb{R} \\ & & \Big\downarrow \text{vertical relation} \\ T_\pi M_a \oplus \mathbb{R} \cong \pi^* \xi_a & \xrightarrow[\text{add } \pi^* T\mathbb{S}^4]{} & \pi^*(\xi_a \oplus T\mathbb{S}^4) \end{array}$$

Figure 11: The stable tangent bundle relation used to compute $p_1(M_a)$.

Let u be a generator of $H^4(\mathbb{S}^4;\mathbb{Z})$. The bundle ξ_a satisfies

$$p_1(\xi_a) = 2au.$$

Since $T\mathbb{S}^4$ is stably trivial, $p_1(T\mathbb{S}^4) = 0$. Hence

$$p_1(M_a) = \pi^* p_1(\xi_a) = 2a\, \pi^*(u) \in H^4(M_a;\mathbb{Z}).$$

This class is nonzero because $a \neq 0$ and $\pi^* : H^4(\mathbb{S}^4) \to H^4(M_a)$ is an isomorphism. On the other hand,

$$p_1(\mathbb{S}^4 \times \mathbb{S}^5) = 0.$$

By Novikov's theorem, rational Pontryagin classes are topological invariants. Therefore M_a cannot be homeomorphic to $\mathbb{S}^4 \times \mathbb{S}^5$.

Theorem 4.3. *For any nonzero $a \in 24\mathbb{Z}$, the manifold*

$$M_a = S(\xi_a)$$

is a smooth closed 9*-manifold such that*

$$M_a \simeq \mathbb{S}^4 \times \mathbb{S}^5, \qquad M_a \not\cong_{\text{homeo.}} \mathbb{S}^4 \times \mathbb{S}^5.$$

Thus M_a *gives a simply connected counterexample to the Hurewicz conjecture.*

Example 4.4. Taking $a = 24$ gives $p_1(M_{24}) = 48\pi^* u \neq 0$, while $M_{24} \simeq \mathbb{S}^4 \times \mathbb{S}^5$.

5 Plumbing and Exotic Spheres

Plumbing is a geometric operation for building new manifolds by gluing disk bundles. In this section we use it in two ways: first to produce the Kervaire spheres (which are exotic in certain dimensions), and then to produce the E_8 plumbing whose boundary generates the subgroup of homotopy spheres bounding parallelizable manifolds.

Local picture of plumbing. Let $D(\xi_i) \to B_i$ be two disk bundles. Choose trivializing patches

$$D^p_{\text{base}} \times D^q_{\text{fiber}} \subset D(\xi_1), \qquad D^q_{\text{base}} \times D^p_{\text{fiber}} \subset D(\xi_2).$$

The local plumbing identifies these two patches by interchanging base and fiber coordinates,

$$(x, y) \longleftrightarrow (y, x).$$

Choose the signs of the coordinate identifications compatibly with the orientations, and round the resulting corners. Thus the zero section of one bundle meets the zero section of the other bundle transversely in one point.

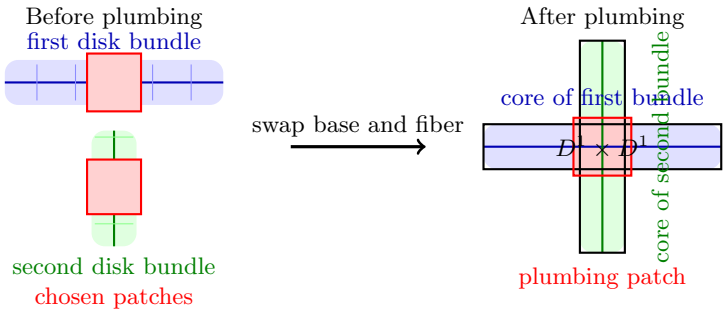


Figure 12: Local picture of plumbing two disk bundles: the selected patches are identified by the coordinate switch $(x, y) \sim (y, x)$.

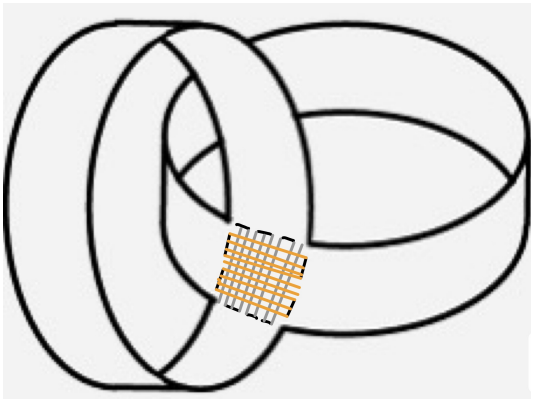


Figure 13: Plumbing disk bundles in $\mathbb{R}^3$[Caf].

5.1 Plumbing two tangent disk bundles

We write $\square$ for plumbing. The simplest family used in Kervaire's construction is

$$M^{2n} = D(T\mathbb{S}^n)\square D(T\mathbb{S}^n).$$

The two zero sections survive as core spheres, so

$$M^{2n} \simeq \mathbb{S}^n \vee \mathbb{S}^n.$$

For $n = 1$ this is the local surface picture above, and $\partial M^2 \cong \mathbb{S}^1$.

Lemma 5.1. *For $n \geq 3$, the boundary is homeomorphic to a sphere if and only if n is odd. In particular, for odd n,*

$$\partial M^{2n} \underset{\text{homeo.}}{\cong} \mathbb{S}^{2n-1}.$$

Proof. The plumbing has a handle decomposition with one 0-handle and two n-handles. Turning this decomposition upside down gives a relative handle decomposition of $(M, \partial M)$ with handles of indices n and $2n$. For $n \geq 3$, these handles do not change the fundamental group. Since $M \simeq S^n \vee S^n$, it follows that

$$\pi_1(\partial M) \xrightarrow{\cong} \pi_1(M) = 0.$$

It remains to compute homology. The long exact sequence for the pair $(M, \partial M)$ reduces to

$$0 \longrightarrow H_n(\partial M) \longrightarrow H_n(M) \xrightarrow{i_*} H_n(M, \partial M) \longrightarrow H_{n-1}(\partial M) \longrightarrow 0.$$

By Poincare–Lefschetz duality,

$$H_n(M, \partial M) \cong H^n(M) \cong \mathrm{Hom}(H_n(M), \mathbb{Z}).$$

If $\{e_1, e_2\}$ is the basis of $H_n(M) \cong \mathbb{Z}^2$ given by the two core spheres, then i_* is represented by the intersection matrix

$$(e_i \cdot e_j) = \begin{cases} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, & n \text{ odd}, \\ \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, & n \text{ even}. \end{cases}$$

Here the diagonal entries are the Euler numbers $\langle e(T\mathbb{S}^n), [\mathbb{S}^n] \rangle = \chi(\mathbb{S}^n)$. When n is odd the matrix is unimodular, hence i_* is an isomorphism and

$$H_n(\partial M) = H_{n-1}(\partial M) = 0.$$

The remaining part of the long exact sequence gives $H_0(\partial M) \cong H_{2n-1}(\partial M) \cong \mathbb{Z}$, with all other homology groups zero. Thus ∂M is a simply connected homology sphere, so the generalized Poincare conjecture gives the desired homeomorphism. If n is even, the matrix has determinant 3 and Smith normal form $\mathrm{diag}(1, 3)$, so $H_{n-1}(\partial M) \cong \mathbb{Z}/3$. Hence the

boundary is not a homology sphere in that case. □

Example 5.2. For $n = 1, 3, 7$, the tangent bundle $T\mathbb{S}^n$ is trivial. In these cases

$$M^{2n} \cong \mathbb{S}^n \times \mathbb{S}^n - \mathrm{int}(\mathbb{D}^{2n}),$$

and therefore $\partial M^{2n} \cong_{\text{diffeo.}} \mathbb{S}^{2n-1}$.

Theorem 5.3 (Kervaire, 1960)**.** *For $n = 5$, the boundary*

$$\partial M^{10} = \Sigma^9$$

is homeomorphic but not diffeomorphic to $\mathbb{S}^9$. Moreover, after capping it topologically by a disk, the resulting topological manifold is not homotopy equivalent to any closed smooth manifold.

Kervaire's original paper is *A manifold which does not admit any differentiable structure*[Ker60]. The obstruction appearing here is the beginning of the Kervaire invariant one problem.

5.2 The Kervaire invariant one problem

Put $n = 2k + 1$. Then the plumbing above has dimension $4k + 2$ and boundary a homotopy sphere

$$\partial M^{4k+2} = \Sigma^{4k+1}.$$

The question whether this homotopy sphere is the standard sphere is controlled by a framed cobordism invariant

$$K(V) \in \mathbb{Z}/2,$$

defined for a closed framed manifold V^{4k+2} and called the Kervaire invariant. The plumbing itself has boundary, so one must distinguish it from such a closed framed manifold. Browder related this invariant to stable homotopy groups of spheres; in modern language the problem asks in which dimensions there exists a framed manifold of Kervaire invariant one.

Theorem 5.4 (Hill–Hopkins–Ravenel)**.** *A smooth framed manifold of Kervaire invariant one can exist only in dimensions*

$$2, 6, 14, 30, 62, 126.$$

The theorem of Hill–Hopkins–Ravenel rules out all other dimensions[HHR16]. Lin–Wang–Xu prove existence in dimension 126 in *On the Last Kervaire Invariant Problem* (arXiv:2412.10879, revised 2025). Thus existence holds precisely in the six displayed dimensions. Their computational methods are developed in the companion work [LWX25a]. For the two-bundle plumbing this means that its boundary is standard precisely when $4k + 2$ is one of these dimensions, or equivalently

$$k \in \{0, 1, 3, 7, 15, 31\}.$$

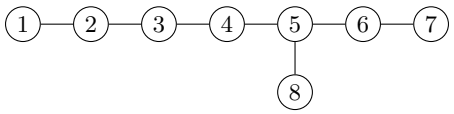
For $k \geq 1$ the boundary class generates bP_{4k+2}, which is either trivial or cyclic of order two. In particular, two copies of this boundary always have standard connected sum.

5.3 The E_8 plumbing

There is a second important plumbing construction. Let $n = 2k$, with $k \geq 1$, and plumb eight copies of $D(T\mathbb{S}^{2k})$ according to the Dynkin diagram E_8. Denote the resulting $4k$-manifold by

$$M_E^{4k}.$$

Here is the labeling used by the matrix below:



The eight zero sections give

$$M_E^{4k} \simeq \bigvee_8 \mathbb{S}^{2k}, \qquad H_{2k}(M_E) \cong \mathbb{Z}^8.$$

The same relative homology sequence gives

$$0 \longrightarrow H_{2k}(\partial M_E) \longrightarrow H_{2k}(M_E) \xrightarrow{i_*} H_{2k}(M_E, \partial M_E) \longrightarrow H_{2k-1}(\partial M_E) \longrightarrow 0.$$

Under the basis given by the eight core spheres and the dual basis of $H^{2k}(M_E)$, the map i_* is represented by the positive E_8 intersection matrix

$$A_{E_8} = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 2 \end{pmatrix}.$$

This matrix satisfies

$$\det(A_{E_8}) = 1, \qquad A_{E_8} \text{ is positive definite}, \qquad \operatorname{sign}(A_{E_8}) = 8.$$

The leading principal minors are $2, 3, 4, 5, 6, 7, 8, 1$, so Sylvester's criterion proves positive definiteness and the last minor proves unimodularity. Therefore ∂M_E is a homology sphere. For $k \geq 2$, the relative handle argument above, now with eight $2k$-handles, proves that it is simply connected, hence a homotopy sphere.

The tangent bundle of the plumbing is stably trivial: on each core sphere its restriction is $TS^{2k} \oplus TS^{2k}$, which is stably trivial. The cores form a wedge, so these trivializations extend to a stable trivialization over M_E. Since this wedge has dimension $2k < 4k$, cancellation in the stable range makes TM_E itself trivial. Thus M_E is parallelizable.

Theorem 5.5 (Kervaire–Milnor signature divisibility)**.** *Let X^{4k} be a closed smooth almost parallelizable manifold. Then*

$$a_k b_k j_k \mid \operatorname{sign}(X),$$

where

$$a_k = \frac{2^{2k}(2^{2k-1}-1)B_k}{(2k)!}, \qquad b_k = \begin{cases} 2(2k-1)!, & k \text{ odd}, \\ (2k-1)!, & k \text{ even}, \end{cases}$$

and

$$j_k = \operatorname{den}\left(\frac{B_k}{4k}\right).$$

Here $B_k = |B_{2k}^{\text{classical}}|$ is the positive Bernoulli number (so $B_1 = 1/6$, $B_2 = 1/30$), determined by

$$\frac{z}{e^z - 1} = 1 - \frac{z}{2} + \frac{B_1}{2!}z^2 - \frac{B_2}{4!}z^4 + \cdots + (-1)^{k-1}\frac{B_k}{(2k)!}z^{2k} + \cdots.$$

For $k = 1$, $a_1 b_1 j_1 = (1/3) \cdot 2 \cdot 24 = 16$. Every closed connected smooth spin 4-manifold is almost parallelizable: on the complement of a disk, $w_2 = 0$ removes the obstruction over the 2-skeleton, $\pi_2(SO) = 0$, and there is no top-dimensional obstruction. Thus this case recovers Rohlin's signature divisibility theorem.

Now suppose that the E_8 boundary were diffeomorphic to the standard sphere. Capping M_E^{4k} by a disk would give a closed smooth almost parallelizable manifold X^{4k} with

$$\operatorname{sign}(X) = \operatorname{sign}(M_E) = 8.$$

For every $k \geq 2$, the divisor $a_k b_k j_k$ is larger than 8, making this impossible. In particular, for $k = 2$,

$$a_2 = \frac{7}{45}, \qquad b_2 = 3!, \qquad j_2 = \operatorname{den}\left(\frac{1}{240}\right) = 240,$$

so

$$\frac{a_2 b_2 j_2}{8} = 28.$$

Thus the boundary of the E_8 plumbing in dimension 8 is an exotic 7-sphere, and its connected-sum powers generate

$$bP_8 \cong \Theta_7 \cong \mathbb{Z}/28.$$

Define bP_{4k} to be the subgroup of oriented homotopy $(4k-1)$-spheres bounding parallelizable $4k$-manifolds. For $k \geq 2$, the full Kervaire–Milnor classification states that the boundary of M_E^{4k} generates this cyclic subgroup

$$bP_{4k} \subset \Theta_{4k-1}, \qquad |bP_{4k}| = \frac{a_k b_k j_k}{8}.$$

The divisibility argument alone gives a lower bound for the order: if the boundary connected sum of ℓ copies has standard boundary, capping it gives signature 8ℓ, so $(a_k b_k j_k/8) \mid \ell$. The assertion that this lower bound is attained uses the additional existence part of the Kervaire–Milnor theorem.

5.4 Connected sum and the group of homotopy spheres

For $n \geq 5$, let Θ_n denote oriented homotopy n-spheres up to orientation-preserving diffeomorphism, equivalently up to oriented h-cobordism. Connected sum defines an abelian group with identity $[S^n]$ and inverse $-[\Sigma] = [-\Sigma]$.

To justify the inverse, put $A = \Sigma \setminus \operatorname{int} D^n$. Van Kampen and Mayer–Vietoris show that A is simply connected and acyclic, hence contractible. After rounding corners, $W = A \times [0,1]$ is a compact contractible $(n+1)$-manifold with boundary $\Sigma \# (-\Sigma)$. Remove an interior $(n+1)$-disk. The remaining cobordism has simply connected boundary components and both boundary inclusions are homology equivalences, hence homotopy equivalences. The h-cobordism theorem applies because $n + 1 \geq 6$, giving

$$\Sigma \# (-\Sigma) \cong_{\text{diffeo.}} S^n.$$

More generally, this argument proves: a closed simply connected smooth n-manifold, $n \geq 5$, bounds a smooth contractible manifold if and only if it is diffeomorphic to S^n.

5.5 The four-dimensional E_8 warning

When $k = 1$, the same E_8 plumbing gives a compact 4-manifold with boundary the Poincare homology sphere. Its fundamental group is the binary icosahedral group I^* of order 120, fitting into $1 \to \{\pm 1\} \to I^* \to A_5 \to 1$ under the double covering $SU(2) \to SO(3)$. Thus it is not a homotopy sphere. Freedman's theory gives a topological contractible 4-manifold with the same boundary. Gluing them produces a closed simply connected topological 4-manifold with E_8 intersection form and signature 8.

Choose opposite boundary orientations when gluing, so that the resulting intersection form is positive E_8. This manifold cannot be smoothed. One way to see this is Rohlin's theorem, since a smooth spin 4-manifold must have signature divisible by 16. Another is Donaldson's theorem: a closed smooth simply connected positive definite 4-manifold has diagonal intersection form, whereas the E_8 form is positive definite, even, unimodular, and not diagonal. The connected sum of two such topological manifolds has form $E_8 \oplus E_8$ and signature 16: Rohlin's divisibility no longer detects it, but Donaldson's theorem still rules out a smoothing. In contrast, adding three copies of $S^2 \times S^2$ gives the form $2E_8 \oplus 3H$, where $H = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. By Freedman's classification this is homeomorphic to a K3 surface with its complex orientation reversed, and is therefore smoothable. With the complex orientation a K3 surface has signature -16, not $+16$.

5.6 Thom-space viewpoint

The next step in the study of differentiable structures is to encode normal bundle data by Thom spaces. For a vector bundle $\xi \to X$, its Thom space is

$$\operatorname{Th}(\xi) = D(\xi)/S(\xi),$$

which, for a compact base X, is the one-point compactification of the total space. The Thom isomorphism identifies

$$H^i(X) \cong \widetilde{H}^{i+r}(\operatorname{Th}(\xi))$$

for an oriented rank r bundle. This is the language used later to pass from explicit plumbing constructions to the classification of smooth structures by normal invariants and surgery.

5.7 Proof of the signature divisibility

Let X^{4k} be closed, oriented and almost parallelizable. Trivializing its tangent bundle off a disk gives a stable clutching class $\alpha \in \pi_{4k-1}(SO) \cong \mathbb{Z}$. If $f : X \to S^{4k}$ collapses the complement of that disk, then f has degree one and the stable tangent bundle is the pullback of the bundle ξ_α over S^{4k}. Consequently its lower Pontryagin classes vanish and

$$\operatorname{sign}(X) = a_k \langle p_k(\xi_\alpha), [S^{4k}] \rangle.$$

Bott integrality gives

$$\operatorname{im}\big(p_k : \widetilde{KO}(S^{4k}) \to \mathbb{Z}\big) = b_k \mathbb{Z}.$$

Indeed, $p_k(\xi) = (-1)^k c_{2k}(\xi \otimes \mathbb{C})$, the image of c_{2k} on $\widetilde{K}(S^{4k})$ is $(2k-1)!\mathbb{Z}$, and complexification on these coefficient groups has image $2\mathbb{Z}$ when k is odd and $\mathbb{Z}$ when k is even. This parity is essential for the factor 2 in b_k.

It remains to prove $J(\alpha) = 0$. Embed X in S^{4k+r} with r sufficiently large. Its stable normal bundle is pulled back from a rank-r bundle $\eta \to S^{4k}$ with stable class $-\alpha$. Pontryagin–Thom collapse gives

$$S^{4k+r} \longrightarrow \operatorname{Th}(\nu_X) \longrightarrow \operatorname{Th}(\eta).$$

The composite sends the fundamental class to the top-dimensional homology generator with degree one. The Thom space has two cells,

$$\operatorname{Th}(\eta) = S^r \cup_{J_r(-\alpha)} D^{r+4k}.$$

After stabilization, its top-cell quotient therefore has a section. The attaching class vanishes, so $J(-\alpha) = 0$, equivalently $J(\alpha) = 0$.

Adams' image-of-J theorem in degree $4k-1$ says that the image is cyclic of order $j_k = \operatorname{den}(B_k/(4k))$. Hence $\ker J = j_k \mathbb{Z}$. With generators chosen compatibly, $\alpha = j_k t$ gives

$$\operatorname{sign}(X) = a_k b_k j_k \, t, \qquad t \in \mathbb{Z},$$

which proves the asserted divisibility.

6 Exotic Spheres and Surgery Theory

The basic idea of surgery is to modify a manifold in its interior while keeping its boundary fixed, so that its homotopy type becomes simpler. The following guiding fact explains why this is useful.

Theorem 6.1. *Let $n \geq 6$, and let W^n be a smooth, compact, contractible manifold whose boundary $M^{n-1} = \partial W$ is simply connected. Then*

$$M^{n-1} = \partial W \underset{\text{diffeo.}}{\cong} \mathbb{S}^{n-1}.$$

Proof. Choose a small embedded disk $\mathbb{D}^n \subset \operatorname{int}(W)$ and set

$$X = W \setminus \operatorname{int}(\mathbb{D}^n).$$

Then X is a cobordism between M and $\mathbb{S}^{n-1}$. Since W is contractible, the inclusions of both boundary components into X are homology equivalences; since the spaces are simply connected, they are homotopy equivalences. Thus X is an h-cobordism. The h-cobordism theorem applies because $\dim M = n - 1 \geq 5$, so X is diffeomorphic to a product cobordism. Hence M is diffeomorphic to $\mathbb{S}^{n-1}$. □

Thus, surgery provides a way to improve the interior of W without changing its boundary $\partial W = M$. If surgery can turn W into a contractible manifold with simply connected boundary, then the boundary must be the standard sphere.

6.1 The Kervaire–Milnor exact sequence

Let Θ_n be the group of oriented homotopy n-spheres up to h-cobordism, with connected sum as the group operation. Let

$$bP_{n+1} \subset \Theta_n$$

be the subgroup consisting of those homotopy spheres which bound a parallelizable $(n+1)$-manifold. The notes lead to the following exact sequence.

Theorem 6.2 (Kervaire–Milnor)**.** *For $n \geq 5$, put $\operatorname{coker} J_n = \pi_n^S / \operatorname{im}(J_n)$. There is an exact sequence*

$$0 \longrightarrow bP_{n+1} \longrightarrow \Theta_n \xrightarrow{\psi} \operatorname{coker} J_n.$$

If $n \not\equiv 2 \pmod 4$, ψ is surjective. If $n \equiv 2 \pmod 4$, its image is the kernel of the Kervaire invariant map

$$K : \operatorname{coker} J_n \longrightarrow \mathbb{Z}/2.$$

No surjectivity of K is asserted in general.

Every homotopy sphere is stably parallelizable. Choose a stable normal framing F on Σ^n. Pontryagin–Thom gives a class $[\Sigma, F] \in \Omega_n^{\text{fr}} \cong \pi_n^S$. Changing the framing changes this class by an element of $\operatorname{im} J_n$, so its coset defines ψ independently of F. A sphere lies in $\ker \psi$ precisely when some framing extends over a parallelizable filling, which explains $\ker \psi = bP_{n+1}$.

For completeness, the Pontryagin–Thom map takes a framed embedding $M^n \subset \mathbb{R}^{n+r}$ to the collapse map

$$S^{n+r} \longrightarrow \mathrm{Th}(\nu_M) \cong M_+ \wedge S^r \longrightarrow S^r.$$

Conversely, the inverse image of a regular value of a smooth map $S^{n+r} \to S^r$ is a framed n-manifold. Transverse homotopies give framed cobordisms, and stabilization makes the constructions inverse. In low degrees,

$$\Omega_0^{\mathrm{fr}} \cong \mathbb{Z}, \quad \Omega_1^{\mathrm{fr}} \cong \mathbb{Z}/2, \quad \Omega_2^{\mathrm{fr}} \cong \mathbb{Z}/2, \quad \Omega_3^{\mathrm{fr}} \cong \mathbb{Z}/24.$$

Theorem 6.3 (Parallelizable fillings)**.** *For $k \geq 2$, the group bP_{4k} is cyclic of order*

$$I_k = 2^{2k-2}(2^{2k-1}-1)\,\mathrm{num}\left(\frac{4|B_{2k}|}{k}\right) = \frac{a_k b_k j_k}{8},$$

where B_{2k} is the classical Bernoulli number and a_k, b_k, j_k use the positive Bernoulli convention of the preceding section. Thus $I_2 = 28$ and $I_3 = 992$. In the high-dimensional range, $bP_m = 0$ for odd m. For $k \geq 1$,

$$bP_{4k+2} \cong \begin{cases} 0, & \text{if a closed framed } (4k+2)\text{-manifold has } K = 1, \\ \mathbb{Z}/2, & \text{if no such manifold exists.} \end{cases}$$

The direction of this equivalence is important: existence of a closed framed manifold with $K = 1$ makes the boundary of the Kervaire plumbing standard. Combining the classical existence results, Hill–Hopkins–Ravenel's nonexistence theorem, and the dimension-126 result of Lin–Wang–Xu, the dimensions admitting $K = 1$ are exactly

$$2,\ 6,\ 14,\ 30,\ 62,\ 126.$$

For $k \geq 1$, therefore, $bP_{4k+2} = 0$ in dimensions $6, 14, 30, 62, 126$, and it is $\mathbb{Z}/2$ in all other dimensions of this form. See On the Last Kervaire Invariant Problem for the final existence case.

6.2 The surgery operation

Let M^n be a smooth manifold, let $0 \leq k \leq n-1$, and suppose that we are given an embedding

$$f : \mathbb{S}^k \times \mathbb{D}^{n-k} \hookrightarrow M^n.$$

The image is required to lie in the interior when M has boundary. The embedding includes a choice of normal framing, not just the assertion that a trivialization exists. In particular, the embedded sphere $f(\mathbb{S}^k \times \{0\})$ has trivial normal bundle. The surgery of index $k+1$ replaces this tubular neighborhood by the complementary handle:

$$M' = \left(M \setminus \operatorname{int} f(\mathbb{S}^k \times \mathbb{D}^{n-k})\right) \cup_{\mathbb{S}^k \times \mathbb{S}^{n-k-1}} \left(\mathbb{D}^{k+1} \times \mathbb{S}^{n-k-1}\right).$$

The two pieces have the same boundary

$$\partial(\mathbb{S}^k \times \mathbb{D}^{n-k}) = \mathbb{S}^k \times \mathbb{S}^{n-k-1} = \partial(\mathbb{D}^{k+1} \times \mathbb{S}^{n-k-1}).$$

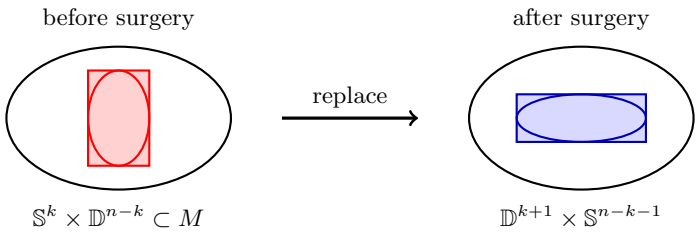


Figure 14: Surgery replaces $\mathbb{S}^k \times \mathbb{D}^{n-k}$ by $\mathbb{D}^{k+1} \times \mathbb{S}^{n-k-1}$ along their common boundary.

The trace of surgery is the cobordism

$$W = (M \times I) \cup_f \left(\mathbb{D}^{k+1} \times \mathbb{D}^{n-k}\right), \qquad \partial W = (-M) \sqcup M'.$$

For closed M, the displayed boundary formula uses the usual incoming/outgoing orientation convention; if M has boundary, the trace also has the fixed side boundary $\partial M \times I$. It has the two homotopy descriptions

$$W \simeq M \cup_{\mathbb{S}^k} \mathbb{D}^{k+1}, \qquad W \simeq M' \cup_{\mathbb{S}^{n-k-1}} \mathbb{D}^{n-k}.$$

The attaching sphere becomes null-homotopic in the trace. The new manifold contains the belt sphere $\{0\} \times S^{n-k-1}$, but it need not represent a nonzero homology class. To identify which classes are killed in M', one must also use the dual handle index.

Proposition 6.4 (Effect on homology)**.** *Suppose that $1 \le k < n-k-1$, and let $\alpha \in H_k(M)$ be represented by the embedded sphere used in the surgery. The relative handle descriptions give*

$$H_i(M') \cong H_i(M) \quad (i < k), \qquad H_k(M') \cong H_k(M)/\langle \alpha \rangle.$$

Indeed, (W, M) has one relative cell of dimension $k+1$, and its homology boundary sends 1 to α. The pair (W, M') has one relative cell of dimension $n-k > k+1$, so $M' \to W$ induces homology isomorphisms through degree k. For $k = 1$, the fundamental group is quotiented by the normal closure of the attaching loop. For $k \ge 2$ in the simply connected setting, the analogous relative homotopy argument kills its homotopy class while preserving lower homotopy groups.

Remark 6.5. In low-dimensional pictures it is tempting to draw the embedded sphere with knots in the ambient Euclidean space. The surgery data, however, lives inside the manifold M itself. For example, a circle that appears knotted in $\mathbb{R}^3$ may be unknotted as a curve on a surface.

6.3 Surgery obstructions for parallelizable manifolds

Let W^d be a compact parallelizable manifold with homotopy-sphere boundary, in dimension $d \ge 6$. All surgeries take place in the interior and preserve the boundary. Fix stable framing data and first kill the fundamental group. Inductively, once lower homotopy groups vanish, Hurewicz identifies the next homotopy group with homology. Below the middle dimension, general position gives embedded representatives, and the stable normal trivialization can be

made an actual normal framing in this range. Choose the surgery framing so that the stable framing extends across the trace. The middle dimension requires additional information.

Filling dimension $d = 4k$, $k \geq 2$. After surgery below the middle dimension, W is $(2k-1)$-connected. Its middle homology is free and carries an even unimodular symmetric intersection form. The simply connected framed surgery obstruction is

$$\frac{\operatorname{sign}(W)}{8} \in L_{4k}(\mathbb{Z}) \cong \mathbb{Z}.$$

A hyperbolic summand has zero obstruction and can be removed by surgery; the positive E_8 form represents a generator. Changing the parallelizable filling of the same boundary changes $\operatorname{sign}(W)/8$ by a multiple of I_k. Thus the boundary invariant is the residue modulo I_k, in agreement with $bP_{4k} \cong \mathbb{Z}/I_k$.

Filling dimension $d = 4k+2$, $k \geq 1$. The middle pairing is skew-symmetric. A compatible quadratic refinement q of its mod-2 reduction is needed in addition to the bilinear pairing. Its Arf invariant is the obstruction in $L_{4k+2}(\mathbb{Z}) \cong \mathbb{Z}/2$.

Odd filling dimension. The simply connected odd-dimensional Wall groups vanish. After the necessary lower-dimensional surgeries, there is no remaining middle-dimensional obstruction. This gives the vanishing of bP_d for odd d in the high-dimensional range.

Do not confuse the dimension of a filling with that of a closed framed representative of π_n^S. For a closed framed $4k$-manifold V, all Pontryagin classes vanish, so Hirzebruch's formula gives $\operatorname{sign}(V) = 0$. For a closed framed $(4k+2)$-manifold, the possible obstruction is its Kervaire invariant. These facts explain the image of ψ described above.

6.4 The Arf invariant and odd-dimensional core spheres

Let V be a free abelian group with a unimodular skew-symmetric pairing λ. It has even rank $2r$ and a symplectic basis $e_1, f_1, \ldots, e_r, f_r$, with $\lambda(e_i, f_j) = \delta_{ij}$ and all other pairings within the two halves zero. A quadratic refinement is a map $q : V/2V \to \mathbb{Z}/2$ satisfying

$$q(x+y) = q(x) + q(y) + \lambda(x,y) \pmod 2.$$

Its Arf invariant is

$$\operatorname{Arf}(q) = \sum_{i=1}^{r} q(e_i)q(f_i) \in \mathbb{Z}/2.$$

It is independent of the symplectic basis. For a fixed rank, rank together with Arf invariant classifies these nonsingular quadratic refinements; under stabilization by hyperbolic planes, Arf invariant alone is complete. In particular, $\operatorname{Arf}(q) = 0$ if and only if there is a symplectic basis with $q(e_i) = q(f_i) = 0$ for every i.

Now let $n \geq 3$ be odd and plumb eight copies of $D(TS^n)$ using the E_8 tree. Call the resulting $2n$-manifold P. Its core classes $a_1, \ldots, a_8$ use the numbering of the preceding section. The intersection matrix is skew-symmetric: set $\lambda(a_i, a_j) = 1$ when $i < j$ are joined by an edge and $\lambda(a_j, a_i) = -1$. It has determinant one, so ∂P is a simply connected homology sphere.

For odd $n \notin \{1, 3, 7\}$, the normal-bundle obstruction for an embedded middle sphere takes values in the order-two kernel generated by $[TS^n]$:

$$\ker\big(\pi_{n-1}SO(n) \to \pi_{n-1}SO\big) = \langle [TS^n] \rangle \cong \mathbb{Z}/2.$$

The corresponding quadratic refinement has $q(a_i) = 1$. Over $\mathbb{Z}/2$, a symplectic basis is

$$\begin{aligned} e_1 &= a_1, & f_1 &= a_2, \\ e_2 &= a_6, & f_2 &= a_7, \\ e_3 &= a_1 + a_3, & f_3 &= a_4, \\ e_4 &= a_1 + a_3 + a_5 + a_7, & f_4 &= a_8. \end{aligned}$$

The supports of e_3 and e_4 contain no tree edges, so $q(e_3) = q(e_4) = 0$. Hence

$$\mathrm{Arf}(q) = 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 1 + 0 \cdot 1 = 0.$$

A hyperbolic basis with vanishing normal obstructions can therefore be realized geometrically. The Whitney trick removes unwanted intersections, and framed surgery on a Lagrangian kills the middle homology while preserving the boundary. The resulting manifold is simply connected and acyclic, hence contractible. The initial theorem implies

$$\partial P \cong_{\text{diffeo.}} S^{2n-1}.$$

For $n = 3, 7$, the tangent spheres are parallelizable and the stable-triviality kernel for their rank-n normal bundles is zero. The same geometric hyperbolic-pair argument applies directly; the description $q(a_i) = 1$ by nontrivial normal bundles is not used in these exceptional dimensions.

6.5 Structure sets

Surgery theory is more powerful than the study of exotic spheres alone. Its general goal is to answer questions of the following form: given a compact CW-complex X, does there exist a closed manifold M with $M \simeq X$? If yes, how many such manifolds are there?

A closed manifold has Poincare duality, so a general CW-complex must satisfy corresponding duality conditions. A finite Poincaré complex of formal dimension n has an orientation local system $\mathcal{O}$ and a fundamental class $[X] \in H_n(X; \mathcal{O})$ such that cap product gives Poincaré duality for every local coefficient system. This is necessary, but not sufficient, for a finite complex to be homotopy equivalent to a closed manifold.

Definition 6.6 (Structure set)**.** Let X^n be a closed connected oriented smooth manifold, $n \geq 5$. Use degree-one homotopy equivalences and orientation-preserving diffeomorphisms. The marked smooth structure set of X is

$$\mathcal{S}(X) = \{(M^n, f) \mid M^n \text{ closed smooth, } f : M \to X \text{ a homotopy equivalence}\} \,/\, \sim,$$

where $(M_0, f_0) \sim (M_1, f_1)$ if there is a diffeomorphism $h : M_0 \to M_1$ such that

$$f_1 \circ h \simeq f_0.$$

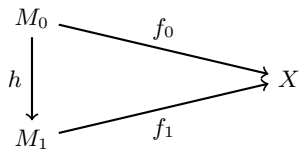


Figure 15: Equivalence in the structure set.

The group of homotopy classes of self-homotopy equivalences acts by postcomposition: $[u] \cdot [M, f] = [M, u \circ f]$. For a degree-minus-one u, reverse the orientation on the source M in this formula so that the new marking again has degree one. If orientations are retained, use only the degree-one subgroup. Forgetting both the marking and orientation gives the set of unoriented diffeomorphism classes

$$\mu(X) = \mathcal{S}(X)/\operatorname{hAut}(X),$$

where $\operatorname{hAut}(X)$ is the group of homotopy automorphisms of X. For example, if $n \geq 5$, then

$$\mathcal{S}(\mathbb{S}^n) \cong \Theta_n,$$

with the degree-one convention, because degree-one maps from a homotopy sphere to S^n are homotopic. If orientation is forgotten, the quotient identifies $[\Sigma]$ with $[-\Sigma]$.

6.6 Normal invariants

A homotopy equivalence carries less information than a diffeomorphism. The missing information is measured by stable normal bundles.

Definition 6.7 (Degree-one normal map)**.** A degree-one normal map to X consists of a degree-one map $f : M^n \to X$, a stable vector bundle ξ over X, and a specified stable bundle isomorphism $b : \nu_M \to f^*\xi$. The normal data includes a compatible identification of the spherical fibration of ξ with the Spivak normal fibration of X. For a manifold target this is the underlying stable spherical fibration of ν_X. Equivalently, one uses a vector-bundle reduction of that fibration, together with its Pontryagin–Thom degree-one normal map.

Definition 6.8 (Normal invariant)**.** Normal maps are normally cobordant when a cobordism over $X \times I$ extends the degree-one maps and their stable normal data. Their normal cobordism classes form $\mathcal{N}(X)$, based at the identity normal map (X, id, ν_X). The bundle identification is part of the data; its existence alone is not a normal invariant.

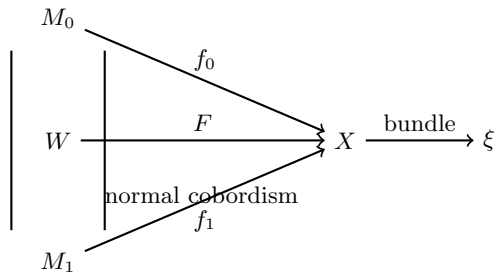


Figure 16: Normal cobordism between two degree-one normal maps.

Let $O = \bigcup_m O(m)$ and let $G = \bigcup_m G(m)$, where $G(m)$ is the monoid of all self-homotopy equivalences of $\mathbb{S}^{m-1}$, of degree ± 1. The stable J-homomorphism comes from $O \to G$. The notation G/O denotes the homotopy fiber of the induced map of classifying spaces $BO \to BG$:

$$G/O \longrightarrow BO \xrightarrow{BJ} BG.$$

$$\begin{array}{ccccc} O(m) & \longrightarrow & O(m+1) & \longrightarrow & O \\ \downarrow{\scriptstyle J} & & \downarrow{\scriptstyle J} & & \downarrow{\scriptstyle J} \\ G(m) & \longrightarrow & G(m+1) & \longrightarrow & G \end{array}$$

Figure 17: Stabilizing the maps $O(m) \to G(m)$ gives the stable J-map; its delooping $BO \to BG$ has homotopy fiber G/O.

Theorem 6.9 (Sullivan)**.** *For a closed smooth manifold X, the normal invariant set is naturally identified with*

$$\mathcal{N}(X) \cong [X, G/O].$$

6.7 The surgery exact sequence

The obstruction groups in simply connected surgery are the Wall L-groups of $\mathbb{Z}$.

Definition 6.10 (Wall L-groups over $\mathbb{Z}$)**.** The simply connected Wall groups are

$$L_n(\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & n \equiv 0 \pmod 4, \\ \mathbb{Z}/2, & n \equiv 2 \pmod 4, \\ 0, & n \text{ odd}. \end{cases}$$

Algebraically, $L_{2k}(\mathbb{Z})$ is the Witt group of nonsingular $(-1)^k$-quadratic forms, with hyperbolic forms representing zero. In degree 0 modulo 4, the underlying symmetric form is even and the integer invariant is signature divided by 8; in degree 2 modulo 4, the invariant is Arf. A bilinear form without its quadratic refinement is insufficient.

Theorem 6.11 (Browder–Novikov–Sullivan–Wall surgery exact sequence)**.** *Let X^n be a closed smooth simply connected manifold with $n \geq 5$. Then there is an exact sequence of*

based sets

$$\cdots \longrightarrow \mathcal{S}(X \times I, \partial) \longrightarrow \mathcal{N}(X \times I, \partial) \xrightarrow{\theta} L_{n+1}(\mathbb{Z})$$
$$\xrightarrow{\omega} \mathcal{S}(X) \xrightarrow{\eta} \mathcal{N}(X) \xrightarrow{\theta} L_n(\mathbb{Z}).$$

The relative terms fix the structures and normal data on both ends of $X \times I$. The group $L_{n+1}(\mathbb{Z})$ acts on $\mathcal{S}(X)$; ω is its orbit map at $[X, \mathrm{id}]$. Exactness means $\operatorname{im}\eta = \theta^{-1}(0)$ and the fibers of η are the orbits of this action. It does not assert that every structure set is naturally a group. In the simply connected case, homotopy equivalences are simple, so no s/h decoration is needed here.

The map

$$\theta : \mathcal{N}(X) \longrightarrow L_n(\mathbb{Z})$$

sends a degree-one normal map (M, f) to the obstruction to doing surgery on f until it becomes a homotopy equivalence. If the obstruction vanishes, then (M, f) can be changed by surgery to represent an element of $\mathcal{S}(X)$.

The obstruction has concrete forms in the three parity cases:

- If $n = 4k$, then
$$\theta(M, f) = \frac{1}{8}\big(\operatorname{sign}(M) - \operatorname{sign}(X)\big) \in \mathbb{Z}.$$
Equivalently, one may compute the signatures using the Hirzebruch L-class.

- If $n = 4k + 2$, then $\theta(M, f)$ is the Arf invariant of the quadratic form on the middle-dimensional surgery kernel after surgery below the middle dimension. This is not generally the invariant of the entire manifold M.

- If n is odd, then $L_n(\mathbb{Z}) = 0$, so there is no simply connected surgery obstruction.

The other maps have the following interpretations.

- The map $\mathcal{S}(X) \to \mathcal{N}(X)$ sends a homotopy equivalence $f : M \to X$ to its normal invariant. For a homotopy inverse $g : X \to M$, one may use $\xi = g^*\nu_M$, with the stable bundle map induced by $g \circ f \simeq \mathrm{id}_M$ and the corresponding spherical normal data.

- The map $L_{n+1}(\mathbb{Z}) \to \mathcal{S}(X)$ is Wall's realization map: an algebraic surgery obstruction is realized geometrically by modifying $X \times I$ and hence changing the resulting manifold structure on X.

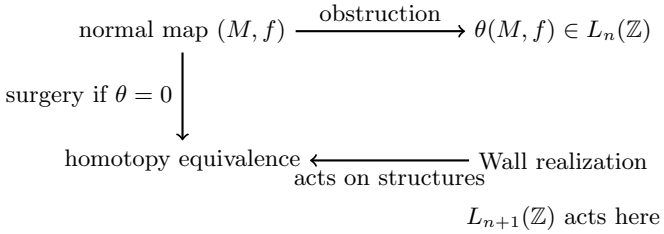


Figure 18: The surgery exact sequence compares normal data, algebraic obstructions, and manifold structures.

Example 6.12 (The case $X = \mathbb{S}^n$)**.** Since $\mathcal{S}(\mathbb{S}^n) \cong \Theta_n$ for $n \geq 5$, the surgery exact sequence recovers the part of the Kervaire–Milnor exact sequence involving

$$\pi_n(G/O), \qquad L_n(\mathbb{Z}), \qquad \Theta_n.$$

This is one reason the classification of exotic spheres is naturally a surgery-theoretic problem.

6.8 The topological version

There is also a topological surgery exact sequence. Let

$$\mathrm{TOP}(m) = \mathrm{Homeo}(\mathbb{R}^m, 0), \qquad \mathrm{TOP} = \bigcup_m \mathrm{TOP}(m).$$

Here homeomorphisms fix the origin, and stabilization is by product with the identity of $\mathbb{R}$. The space G/TOP is the homotopy fiber of

$$B\,\mathrm{TOP} \longrightarrow BG.$$

For a closed topological manifold X, the topological normal invariants are

$$\mathcal{N}^{\mathrm{TOP}}(X) \cong [X, G/\mathrm{TOP}].$$

For closed simply connected topological n-manifolds with $n \geq 5$, the topological surgery exact sequence has the same shape:

$$\begin{aligned} \cdots \longrightarrow \mathcal{S}^{\mathrm{TOP}}(X \times I, \partial) &\longrightarrow \mathcal{N}^{\mathrm{TOP}}(X \times I, \partial) \longrightarrow L_{n+1}(\mathbb{Z}) \\ &\longrightarrow \mathcal{S}^{\mathrm{TOP}}(X) \longrightarrow \mathcal{N}^{\mathrm{TOP}}(X) \longrightarrow L_n(\mathbb{Z}). \end{aligned}$$

The difference between the smooth and topological theories is encoded by the comparison between G/O and G/TOP; this is where Kirby–Siebenmann theory enters.

For spheres, the topological generalized Poincare conjecture says that the topological structure set of $\mathbb{S}^n$ is trivial in high dimensions. Thus the topological theory has fewer exotic objects than the smooth theory.

For $n \geq 5$, the relative topological structure set of $(S^n \times I, \partial)$ is also trivial by topological s-cobordism. The sphere surgery sequence thus identifies

$$\theta : \pi_n(G/\mathrm{TOP}) \xrightarrow{\cong} L_n(\mathbb{Z}).$$

These are the homotopy groups in the range used here; extending the calculation to low dimensions requires additional results.

6.9 Examples from the surgery exact sequence

The fibration $G/O \to BO \to BG$ gives, for $n \geq 2$,

$$0 \longrightarrow \operatorname{coker} J_n \longrightarrow \pi_n(G/O) \longrightarrow \ker J_{n-1} \longrightarrow 0.$$

Thus $\pi_n(G/O)$ must not simply be replaced by $\operatorname{coker} J_n$ in every dimension. Bott periodicity and Adams' theorem show that $\ker J_{n-1}$ is infinite cyclic when $n \equiv 0 \pmod 4$ and is zero in the other positive degrees relevant here.

For S^6, $\pi_6(G/O) \cong \pi_6^S \cong \mathbb{Z}/2$, and θ is the nonzero Arf map to $L_6(\mathbb{Z})$. Since $L_7(\mathbb{Z}) = 0$, exactness gives $\Theta_6 = 0$. For S^7, $\operatorname{coker} J_7 = 0$ and $\pi_7(G/O) = 0$. The image of

$$\pi_8(G/O) \xrightarrow{\theta} L_8(\mathbb{Z}) \cong \mathbb{Z}$$

is $28\mathbb{Z}$, so Wall realization gives $\Theta_7 \cong \mathbb{Z}/28$, consistently with the E_8 plumbing and Eells–Kuiper calculation.

The lecture's projective-space example is a further application. For $m \geq 3$, the cofiber sequence

$$\mathbb{CP}^{m-1} \longrightarrow \mathbb{CP}^m \longrightarrow S^{2m}$$

combines with topological surgery and the splitting invariants of the standard filtration by projective subspaces. These give a parametrization

$$\mathcal{S}^{\mathrm{TOP}}(\mathbb{CP}^m) \longleftrightarrow \bigoplus_{i=1}^{m-1} L_{2i}(\mathbb{Z}).$$

This parametrization uses Sullivan's splitting theorem (see [Wal99, Chapter 14C]), not merely the cofiber sequence or a formal replacement of maps into G/TOP by cohomology. Integer coordinates occur for even i and mod-2 coordinates for odd i. In particular there are infinitely many marked structures. A self-homotopy equivalence acts on H^2 by $x \mapsto \pm x$; these are the two homotopy classes of self-equivalences of $\mathbb{CP}^m$. Quotienting by this finite action still leaves infinitely many homeomorphism types homotopy equivalent to $\mathbb{CP}^m$.

7 Whitehead Torsion and the s-Cobordism Theorem

Whitehead torsion measures the difference between homotopy equivalence and simple homotopy equivalence. For an h-cobordism it is exactly the obstruction to being a product. We develop the algebraic construction first, then apply it to finite CW-complexes and manifolds. For further details and conventions, see [Mne14, Maz63].

7.1 Elementary matrices and algebraic K_1

Let R be an associative ring with identity. Write $GL_m(R)$ for the invertible $m \times m$ matrices and use stabilization $A \mapsto \operatorname{diag}(A, 1)$ to define $GL(R) = \varinjlim_m GL_m(R)$. For $i \neq j$ and $a \in R$, the elementary matrix

$$e_{ij}(a) = I + aE_{ij}, \qquad e_{ij}(a)^{-1} = e_{ij}(-a)$$

performs an elementary row operation by left multiplication. Let $E(R)$ be the stable subgroup generated by these matrices.

Lemma 7.1 (Whitehead's lemma)**.** *The stable elementary subgroup is the commutator subgroup:*

$$E(R) = [GL(R), GL(R)].$$

Proof sketch. For distinct i, j, k, direct multiplication gives

$$e_{ij}(a) = [e_{ik}(a), e_{kj}(1)],$$

so every elementary generator is a commutator after stabilization. In the other direction, block elementary operations show that $\operatorname{diag}(A, A^{-1})$ is elementary. One useful factorization is

$$\begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix} = \begin{pmatrix} I & A \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -A^{-1} & I \end{pmatrix} \begin{pmatrix} I & A \\ 0 & I \end{pmatrix} \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}.$$

Each block factor is a product of elementary matrices. Together with block interchange, this identifies the stable elementary quotient as an abelian group, with $[AB] = [A] + [B]$. Hence every stable commutator is elementary. □

Definition 7.2. The first algebraic K-group is

$$K_1(R) = GL(R)/E(R) = GL(R)_{\mathrm{ab}}.$$

A unital ring map $R \to S$ acts on matrix entries and induces $K_1(R) \to K_1(S)$; thus K_1 is a functor. Also $K_1(R \times S) \cong K_1(R) \oplus K_1(S)$.

For commutative R, determinant gives a split exact sequence

$$0 \longrightarrow SK_1(R) \longrightarrow K_1(R) \xrightarrow{\det} R^\times \longrightarrow 1,$$

where $SK_1(R)$ is its kernel. For a field or a Euclidean domain, row reduction gives $SK_1(R) = 0$. In particular $K_1(F) = F^\times$ for a field and $K_1(\mathbb{Z}) = \{\pm 1\}$. The reduced group is

$$\widetilde{K}_1(R) = K_1(R) / \operatorname{im}\big(K_1(\mathbb{Z}) \to K_1(R)\big).$$

For a noncommutative ring, ordinary determinant is not available, but K_1 still makes sense.

7.2 Group rings and Whitehead groups

For a group π, its integral group ring consists of finite sums $\sum_g a_g g$, with multiplication induced by that of π. It is commutative exactly when π is abelian. The elements $\pm g$ are units, called the trivial units.

Definition 7.3. The Whitehead group is

$$\mathrm{Wh}(\pi) = K_1(\mathbb{Z}[\pi])/\langle [\pm g] : g \in \pi \rangle.$$

The quotient is by their images in K_1, not by the group elements before passing to K_1.

Thus $\mathrm{Wh}(1) = 0$. Another important calculation is $\mathrm{Wh}(\mathbb{Z}^r) = 0$ for $r \geq 0$, a consequence of the fundamental theorem of algebraic K-theory for these Laurent polynomial rings.

For a finite group G, the Whitehead group is finitely generated and

$$\mathrm{Wh}(G) \otimes \mathbb{Q} \cong \mathbb{Q}^{r(G)-q(G)}.$$

Here $r(G)$ counts conjugacy classes of unordered pairs $\{g, g^{-1}\}$, and $q(G)$ counts conjugacy classes of cyclic subgroups. Equivalently, these count irreducible real and rational representations. The formula gives the rank; it does not rule out torsion for an arbitrary finite group. For an odd prime p, the additional integral calculation for cyclic groups gives

$$\mathrm{Wh}(C_p) \cong \mathbb{Z}^{(p-3)/2}.$$

In particular $\mathrm{Wh}(C_5) \cong \mathbb{Z}$.

Let $C_5 = \langle t \mid t^5 = 1 \rangle$. A concrete unit is

$$u = t + t^{-1} - 1, \qquad u^{-1} = t^2 + t^{-2} - 1.$$

Their product is 1 using $t^5 = 1$. The class $[u]$ generates $\mathrm{Wh}(C_5)$. To see directly that it has infinite order, evaluate t at $\zeta = e^{2\pi i/5}$. Then u maps to $2\cos(2\pi/5) - 1$, whose absolute value is neither zero nor one, whereas every trivial unit has absolute value one. This proves infinite order; the generator assertion uses the integral unit-group calculation.

The assertion $\mathrm{Wh}(G) = 0$ for every torsion-free group is a conjecture in general, and follows from the algebraic K-theory Farrell–Jones conjecture. It is known for many classes of groups; see the 2025 survey [Lüc25]. It must not be used without hypotheses as a general theorem.

7.3 Torsion of a based chain complex

A finite based free chain complex over R is a finite sequence of finitely generated free modules with ordered bases,

$$\cdots \longrightarrow C_{i+1} \xrightarrow{d_{i+1}} C_i \xrightarrow{d_i} C_{i-1} \longrightarrow \cdots, \qquad d_i d_{i+1} = 0.$$

Its homology is $H_i(C) = \ker d_i / \operatorname{im} d_{i+1}$. Such a bounded acyclic complex is contractible: there is a degree-one map h with

$$dh + hd = 1.$$

Indeed, split the surjection onto the lowest nonzero free module and induct upward; the successive kernels are projective. Conversely this identity immediately implies acyclicity. The bounded projective hypothesis matters here; arbitrary acyclic complexes need not be contractible.

Choose a contraction with $h^2 = 0$, which can be arranged using the splittings. Then $(d + h)^2 = 1$, and hence

$$d + h : C_{\text{odd}} = \bigoplus_i C_{2i+1} \xrightarrow{\cong} C_{\text{even}} = \bigoplus_i C_{2i}.$$

For the rings used below, in particular $\mathbb{Z}[\pi]$, the two free modules have the same rank. With the induced ordered bases this is an invertible matrix A.

Definition 7.4 (Chain torsion)**.** We use the odd-to-even convention

$$\tau(C) = [A] \in K_1(R).$$

It is independent of the contraction. Changes of bases have a controlled effect, so the bases are essential; after quotienting signs the class lies in $\widetilde{K}_1(R)$.

For the two-term complex $0 \to R^m \xrightarrow{A} R^m \to 0$ in degrees $1, 0$, its torsion is $[A]$. Reversing the degrees reverses its sign in additive K_1 notation.

7.4 Cellular chains and local coefficients

A CW-complex is built by attaching disks to successive skeleta; its topology is the weak topology with respect to the characteristic maps, and it satisfies closure-finiteness. For example,

$$S^n = e^0 \cup e^n, \qquad \mathbb{CP}^m = \mathbb{CP}^{m-1} \cup e^{2m}.$$

A compact smooth manifold admits a finite triangulation. General topological manifolds have CW homotopy type, but one must not infer that every topological manifold admits a triangulation.

For a connected finite CW-complex X with $\pi = \pi_1(X)$, choose an orientation and a lift to the universal cover for every cell. Deck transformations make $C_*(\widetilde{X})$ a based free left $\Lambda = \mathbb{Z}[\pi]$-complex, with one basis vector for each cell of X. Changing a lift or orientation multiplies a basis element by a unit $\pm g$.

Example 7.5 (The circle)**.** With one vertex and one edge, $C_*(S^1; \mathbb{Z})$ has differential zero. Its universal cover gives the different complex

$$0 \longrightarrow \mathbb{Z}[t, t^{-1}] \xrightarrow{t-1} \mathbb{Z}[t, t^{-1}] \longrightarrow 0.$$

Its homology is $H_1 = 0$ and $H_0 \cong \mathbb{Z}$, with trivial t-action, as expected for the contractible universal cover.

If A is a right Λ-module and B is a left Λ-module, define

$$H_*(X;A) = H_*(A \otimes_\Lambda C_*(\widetilde{X})),$$
$$H^*(X;B) = H^*(\mathrm{Hom}_\Lambda(C_*(\widetilde{X}), B)).$$

In particular $H_*(X;\Lambda) = H_*(\widetilde{X})$, with the appropriate regular-module convention. For a locally finite lifted cell structure, $H^*(X;\Lambda)$ identifies with compactly supported cohomology of $\widetilde{X}$, not in general ordinary cohomology.

For $\mathbb{RP}^2$, put $\Lambda = \mathbb{Z}[C_2]$ with generator t. The lifted cellular complex has differentials $d_2 = 1 + t$ and $d_1 = t - 1$. For the sign local system $\mathbb{Z}_-$, where t acts by -1, the tensor complex is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{-2} \mathbb{Z} \longrightarrow 0.$$

Thus $H_2(\mathbb{RP}^2;\mathbb{Z}_-) = \mathbb{Z}$, $H_1 = 0$, and $H_0 = \mathbb{Z}/2$, illustrating the orientation local system.

7.5 Whitehead torsion and simple homotopy equivalence

Let $K \subset L$ be a finite CW-pair and suppose the inclusion is a homotopy equivalence. Using the induced identification of fundamental groups, the relative lifted complex $C_*(\widetilde{L}, \widetilde{K})$ is finite, based, free and acyclic. Define

$$\tau(L,K) = \tau\big(C_*(\widetilde{L}, \widetilde{K})\big) \in \mathrm{Wh}(\pi_1 L).$$

The quotient by $[\pm g]$ removes the choices of cell lifts and orientations; permutations contribute only signs. Changing basepoint changes the group identification by an inner automorphism, which acts trivially on the Whitehead group.

An elementary collapse removes a cell together with a free face. The relative complex of this elementary pair has differential a trivial unit, hence zero Whitehead torsion. An elementary expansion is its inverse.

Theorem 7.6 (Whitehead)**.** *A homotopy equivalence of finite CW-complexes is simple if and only if its Whitehead torsion is zero. Equivalently, it is homotopic to a finite composition of elementary expansions and collapses. Both operations are allowed; a collapse-only sequence is too restrictive.*

For a cellular homotopy equivalence $f : X \to Y$, its mapping cylinder

$$M_f = Y \cup_f (X \times I)$$

contains the incoming copy of X and retracts onto Y. Define

$$\tau(f) = \tau(M_f, X) \in \mathrm{Wh}(\pi_1 Y),$$

using the retraction to identify fundamental groups. Cellular approximation and homotopy invariance extend the definition to any homotopy equivalence between finite CW-complexes. In particular $\tau(\mathrm{id}) = 0$, and every homotopy equivalence between simply connected finite CW-complexes is simple because $\mathrm{Wh}(1) = 0$.

7.6 Properties of torsion

For homotopy equivalences $f : X \to Y$ and $g : Y \to Z$, the composition formula is

$$\tau(g \circ f) = \tau(g) + g_*\tau(f) \in \mathrm{Wh}(\pi_1 Z).$$

Homotopy of the map and subdivision leave torsion unchanged. If $X = X_1 \cup X_2$, $Y = Y_1 \cup Y_2$ are finite CW decompositions with connected intersections, and f restricts to homotopy equivalences $f_i : X_i \to Y_i$ for $i = 0, 1, 2$, where $X_0 = X_1 \cap X_2$, then

$$\tau(f) = (j_1)_*\tau(f_1) + (j_2)_*\tau(f_2) - (j_0)_*\tau(f_0),$$

where $j_i : Y_i \hookrightarrow Y$. All terms must be transported to the same Whitehead group.

For homotopy equivalences $f : X \to X'$ and $g : Y \to Y'$, let $i : X' \to X' \times Y'$ and $j : Y' \to X' \times Y'$ be the factor inclusions at chosen basepoints. Then

$$\tau(f \times g) = \chi(Y)\, i_*\tau(f) + \chi(X)\, j_*\tau(g).$$

In particular $f \times \mathrm{id}_{S^1}$ is always simple because $\chi(S^1) = 0$.

Theorem 7.7 (Chapman)**.** *Every homeomorphism of finite CW-complexes is a simple homotopy equivalence. In particular, its Whitehead torsion vanishes.*

For smooth manifolds one may compute torsion using compatible finite triangulations, and subdivision invariance removes that choice.

7.7 The s-cobordism theorem

For a smooth h-cobordism $(W^{n+1}; M^n, N^n)$, set

$$\tau(W, M) = \tau(M \hookrightarrow W) \in \mathrm{Wh}(\pi_1 W),$$

and transport it to $\mathrm{Wh}(\pi_1 M)$ using the inclusion isomorphism.

Theorem 7.8 (s-cobordism: Barden–Mazur–Stallings)**.** *Let M^n be a closed connected smooth manifold with $n \geq 5$. Torsion gives a bijection*

$$\left\{ \begin{array}{l} \textit{smooth } h\textit{-cobordisms starting at } M \\ \textit{up to diffeomorphism relative to } M \end{array} \right\} \longleftrightarrow \mathrm{Wh}(\pi_1 M).$$

The zero element corresponds to $M \times I$. In particular, W is a product relative to M if and only if $\tau(W, M) = 0$.

The relative qualification is essential: this classifies cobordisms with their incoming boundary identified with M, not merely their outgoing manifolds. When $\pi_1 M = 1$, it recovers the simply connected h-cobordism theorem.

Example 7.9. Let $M = S^{2r-1}/C_5$ be a lens space obtained from the standard free scalar action, with $2r - 1 \geq 5$. Since $\mathrm{Wh}(C_5) \cong \mathbb{Z}$, there are infinitely many h-cobordisms over M up to diffeomorphism relative to M. This alone does not assert that all outgoing boundary manifolds are pairwise nondiffeomorphic.

Corollary 7.10. *Let $(W; M, N)$ be a smooth h-cobordism and let X be a closed connected smooth manifold with $\chi(X) = 0$. If $\dim(W \times X) \geq 6$, then $W \times X$ is a product cobordism relative to $M \times X$, and $M \times X \cong N \times X$.*

Proof. The relative product formula gives $\tau(W \times X, M \times X) = \chi(X)\, i_*\tau(W, M) = 0$. Apply the s-cobordism theorem. In particular this applies to $X = S^1$ whenever the dimension condition holds. □

7.8 Duality and open products

The orientation character $w : \pi_1 W \to \{\pm 1\}$ defines an anti-involution on the group ring,

$$\overline{\sum_g a_g g} = \sum_g a_g w(g) g^{-1}.$$

It satisfies $\overline{ab} = \bar{b}\bar{a}$. Transpose followed by this entrywise involution sends an invertible matrix A to $A^* = (\bar{a}_{ji})$ and induces an involution on $\mathrm{Wh}(\pi_1 W)$.

Theorem 7.11 (Duality)**.** *For an h-cobordism $(W^{n+1}; M^n, N^n)$,*

$$\tau(W, N) = (-1)^n \overline{\tau(W, M)} \quad \textit{in } \mathrm{Wh}(\pi_1 W).$$

This follows by comparing the relative handle complexes from the two ends, which are dual complexes in complementary dimensions. The exponent is the boundary dimension n, and the involution includes the orientation character.

Theorem 7.12 (Stallings' open-product consequence)**.** *If $(W^{n+1}; M, N)$ is a smooth h-cobordism with $n \geq 5$, then*

$$W \setminus N \cong M \times [0, \infty), \qquad \operatorname{int} W \cong M \times \mathbb{R} \cong N \times \mathbb{R}.$$

No simple-connectivity assumption is required.

One proof chooses an inverse h-cobordism, using realization of the negative torsion, so that consecutive composites are products by s-cobordism. Stacking alternating copies gives an infinite collar; regrouping the same stack from either end yields the displayed open-product diffeomorphisms. This is the infinite cancellation argument, usually called the Mazur swindle. It does not imply that the original compact cobordism has zero torsion.

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